

210A Algebra Notes

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Discussion 0 - Group Theory Preview

Definition 1. A group is a set G together with a binary operation $\cdot : G \times G \rightarrow G$ (the fact that $\forall a, b \in G, a \cdot b \in G$ is called "closure" property) satisfying the following properties:

- (i) (Associativity) $\forall a, b, c \in G, a \cdot (b \cdot c) = (a \cdot b) \cdot c$
- (ii) (Identity) $\exists e \in G, a \cdot e = e \cdot a = a$
- (iii) (Inverse) $\forall a \in G, \exists b \in G$ such that $a \cdot b = b \cdot a = e$

Exercise 1. Identity and inverses are unique. That is, if e and f are identities, then $e = f$ (hints: consider ef), and for a fixed $a \in G$, if b and c are inverses, then $b = c$ (hints: consider bac).

Remark. We will denote $a \cdot b$ more shortly as ab unless otherwise specified.

Here's a nice prototypical example of where groups come from:

Example 1 (Prototypical Example). Let X be a set. Take $G = S_X = \{f : X \rightarrow X \mid f \text{ is bijective}\}$, with $f \cdot g = f \circ g$. When $X = [n] = \{1, 2, \dots, n\}$, we call $S_X := S_n$ the symmetric group on n elements. The group elements simply permute the numbers $1, 2, \dots, n$, so sometimes, S_n is also called the permutation group.

More generally, if X is a set with "structure", the set of "automorphisms" of X ("structure"-preserving bijective functions on X) gives rise to groups.

Example 2 (General Linear Groups). Let V be a (finite dimensional) vector space. Then, $G = \text{Gl}(V) = \{T : V \rightarrow V \mid T \text{ is invertible linear}\}$ (together with $\cdot = \circ$) forms a group, called the *general linear group* on V . When $V = \mathbb{R}^n$ or $\mathbb{C}^n$, $\text{Gl}(V) = \text{Gl}(\mathbb{R}^n) =: \text{Gl}(n, \mathbb{R})$, and $\text{Gl}(V) =: \text{Gl}(n, \mathbb{C})$.

The *special linear group* is defined as $\text{Sl}(V) = \{T : V \rightarrow V \mid \det(T) = 1\}$.

Example 3 (Orthogonal Linear Groups). Let V be a vector space now with an inner product $\langle, \rangle$. Then, $\text{O}(V) := \{T : V \rightarrow V \mid \langle Tv, Tw \rangle = \langle v, w \rangle\}$ is called the *orthogonal group*. The special orthogonal group is defined to be $\text{SO}(V) := \text{O}(V) \cap \text{Sl}(V)$.

Example 4 (Complex Conjugation). We know $\mathbb{R} \subset \mathbb{C}$ as fields. We can consider the group

$$G = \text{Aut}_{\mathbb{R}}(\mathbb{C}) = \{\sigma : \mathbb{C} \rightarrow \mathbb{C} \mid \sigma \text{ a field automorphism fixing } \mathbb{R}, (\text{that is, } \forall r \in \mathbb{R}, \sigma(r) = r)\}$$

(Here, field automorphism means σ is an invertible map such that $\sigma(zw) = \sigma(z)\sigma(w)$, $\sigma(z+w) = \sigma(z) + \sigma(w)$, $\sigma(1) = 1$.)

It turns out, $\text{Aut}_{\mathbb{R}}(\mathbb{C}) = \{\text{id}_{\mathbb{C}}, \text{Conj} : a + bi \mapsto a - bi\}$, which corresponds (isomorphically) to $S_{\{i, -i\}}$ (look at where i gets sent). This is an instance of Galois theory and illustrates why Groups are important in studying (extension of) fields. More detail to be taught in 210B in the Galois Theory unit.

Example 5 (Dihedral Groups). For each $n \geq 3$, we can define D_{2n} as follows:

$$D_{2n} = \{T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \mid T \text{ is an isometry } (d(Tx, Ty) = d(x, y)) \text{ where } T(n\text{-Gon}) = n\text{-Gon}\}$$

The last condition means given some fixed n -Gon in the plane, T sends the n -Gon back to itself.

It turns out, D_{2n} is finite and has exactly $2n$ -many elements.

We're ready to tackle one of the qual problems with bare hands.

Adapted From Spring 2020 Problem 6: Let $\mathbb{F}$ be a field with $m \geq 3$ elements. Find $|\text{Sl}_2(\mathbb{F})|$ in terms of m .

Solution. We will first compute $|\text{Gl}_2(\mathbb{F})|$ in terms of m . If $T : \mathbb{F}^2 \rightarrow \mathbb{F}^2$ is linear, then

$$\begin{aligned} T \text{ is invertible} &\iff Te_1, Te_2 \text{ are linearly independent} \\ &\iff Te_1 \neq 0 \text{ and } Te_2 \notin \text{Span}(Te_1) \end{aligned}$$

There are exactly $m^2 - 1$ choices for the first column, and $m^2 - m$ choices for the second column. Hence, there are exactly $(m^2 - 1)(m^2 - m)$ many elements in $\text{Gl}_2(\mathbb{F})$.

Exercise 2. For all $\alpha \in \mathbb{F} - \{0\}$, $|\{T \in \text{Gl}_2(\mathbb{F}) \mid \det(T) = \alpha\}| = |\text{Sl}_2(\mathbb{F})|$.

Hints. Consider $T \mapsto (\alpha Te_1, Te_2)$. This is a bijection. □

Using the exercise, we can conclude that $|\text{Gl}_2(\mathbb{F})| = |\mathbb{F} - \{0\}| \cdot |\text{Sl}_2(\mathbb{F})| = (m-1)|\text{Sl}_2(\mathbb{F})|$. Hence, $|\text{Sl}_2(\mathbb{F})| = \boxed{m^3 - m}$. □

Definition 2 (Conjugate Elements). Given a group G , we say a is conjugate to b (and write $a \sim b$) if and only if $\exists g \in G$ such that $a = bgb^{-1}$.

Exercise 3. Show that $\sim$ is an equivalence relation.

Definition 3 (Conjugacy Class). The conjugacy class of a is defined to be $[a]_{\sim} = \{gag^{-1} \mid g \in G\}$.

Spring 2017 Problem 1: Choose a representative for every conjugacy class in the group $\text{Gl}_2(\mathbb{R})$.

Solution. We analyze the Jordan Normal Form of $A \in \text{Gl}_2(\mathbb{R})$ since A is conjugate to it in the larger group $\text{Gl}_2(\mathbb{C})$. For that, we can analyze the characteristic polynomial $\det(xI - A)$ which is a real quadratic polynomial.

Case 1: $\det(xI - A)$ has no real roots

In this case, the roots are given by $a \pm bi$ for $b \neq 0$, and $a, b \in \mathbb{R}$. Then,

$$A \sim_{\mathbb{C}} \begin{pmatrix} a+bi & 0 \\ 0 & a-bi \end{pmatrix}$$

So there exists a complex basis $u_1 + iv_1, u_2 + iv_2$ (with $u_1, u_2, v_1, v_2 \in \mathbb{R}^2$) so that $A(u_1 + iv_1) = (a + bi)(u_1 + iv_1)$ and $A(u_2 + iv_2) = (a - bi)(u_2 + iv_2)$.

Exercise 4. The u_1, v_1 are linearly independent over $\mathbb{R}$, and with respect to that basis, A has the form $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$

From the exercise 4, it follows that $A \sim_{\mathbb{R}} \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$

Case 2: $\det(xI - A)$ has two distinct real roots.

Call λ_1, λ_2 the two distinct roots of $\det(xI - A)$. Then, $A \sim_{\mathbb{R}} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$.

Case 3: $\det(xI - A)$ has a repeated real root.

Here, $\det(xI - A) = (x - \lambda)^2$ where $\lambda \in \mathbb{R}$ is the repeated root. Then, the generalized eigenspace $G_\lambda = \{v \in \mathbb{R}^2 \mid (A - \lambda I)^2 v = 0\}$ is all of $\mathbb{R}^2$. If $E_\lambda = \{v \in \mathbb{R}^2 \mid (A - \lambda I)v = 0\}$ is all of $\mathbb{R}^2$, then $A = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$. Otherwise, E_λ has dimension 1, and thus, for a given $0 \neq v \in E_\lambda$, $v, (A - \lambda)v$ is a basis of G_λ , so $A \sim_{\mathbb{R}} \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. $\square$

Discussion 1 - Unit Group of a Finite Field, Zorn's Lemma

We will prove the following theorem about the unit group of a finite field.

Theorem 1. Let F be a finite field. Then the group $(F, \times)$ is cyclic.

We will prove this via a series of lemmas.

Lemma 1. Let G be a finite abelian group. If $\text{ord}(a) = m$ and $\text{ord}(n)$, and $\text{gcd}(m, n) = 1$, then $\text{ord}(ab) = mn$.

Proof. Clearly, $(ab)^{mn} = a^{mn}b^{mn} = e \cdot e = e$. Now, suppose $(ab)^K = e$ for some K . Then, $a^K = b^{-K}$. If we raise both sides by n , we get $a^{nK} = b^{-nK} = (b^{-n})^K = e$. Hence, $m = \text{ord}(a) \mid nK$. Since $\text{gcd}(m, n) = 1$, we get $m \mid K$. Similarly, if we raised both sides by m instead, we would get $n \mid K$. Hence, we have $\text{lcm}(m, n) = mn \mid K$. $\square$

Remark. If $\gcd(m, n) \neq 1$, it's not in general true that $\text{ord}(ab) = \text{lcm}(m, n)$. For instance, take any $a \neq e$ and $b = a^{-1}$. Then, $\text{ord}(a) = \text{ord}(b) = m > 1$, but $\text{ord}(ab) = \text{ord}(e) = 1 \neq m = \text{lcm}(m, m)$.

Lemma 2. Let G be a finite abelian group. Set $m = \text{lcm}_{g \in G}(\text{ord}(g))$ (note clearly, that 1) $\forall g \in G, g^m = e$ and 2) since $g^{|G|} = e$ for all $g \in G$, $m \mid |G|$). Then, there exists $g \in G$ such that $\text{ord}(g) = m$.

Proof. Factorize $m = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$. For each $g \in G$, define $e(g)_1, \dots, e(g)_k$ to be such that $\text{ord}(g) = p_1^{e(g)_1} p_2^{e(g)_2} \cdots p_k^{e(g)_k}$. Then, note

$$\begin{aligned} m &= \text{lcm}_{g \in G}(\text{ord}(g)) \\ &= \text{lcm}_{g \in G}(p_1^{e(g)_1} p_2^{e(g)_2} \cdots p_k^{e(g)_k}) \\ &= p_1^{\max(e(g)_1)} p_2^{\max(e(g)_2)} \cdots p_k^{\max(e(g)_k)} = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} \end{aligned}$$

So choose $g_1, g_2, \dots, g_k$ such that $e(g_1)_1 = e_1, e(g_2)_2 = e_2, \dots, e(g_k)_k = e_k$. Then,

$$\begin{aligned} \text{ord}(g_1) &= p_1^{e_1} l_1 \\ \text{ord}(g_2) &= p_2^{e_2} l_2 \\ &\vdots \\ \text{ord}(g_k) &= p_k^{e_k} l_k \end{aligned}$$

Where $p_j \nmid l_j$. Now, if we set $h_1 = g_1^{l_1}, h_2 = g_2^{l_2}, \dots, h_k = g_k^{l_k}$, then $\text{ord}(h_1) = p_1^{e_1}, \text{ord}(h_2) = p_2^{e_2}, \dots, \text{ord}(h_k) = p_k^{e_k}$. By Lemma 1, the element $h = h_1 h_2 \cdots h_k$ has the desired order $p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} = m$. $\square$

Proof of Theorem 1. Set $n = |F^\times| = |F| - 1$, and as before, $m = \text{lcm}_{\alpha \in F^\times}(\text{ord}(\alpha))$. By 2) of Lemma 2 (in parentheses), $m \mid n$ and hence, $m \geq n$. Assume for the sake of contradiction, that $m < n$. Then, by 1) of Lemma 2 (again in parentheses), $\alpha^m = 1$ for all $\alpha \in F^\times$. But then, the polynomial $X^m - 1$ has $n > m = \deg(X^m - 1)$ many roots, which is impossible.

Now, using Lemma 2, choose $\alpha \in F^\times$ such that $\text{ord}(\alpha) = m = n$. Then, α generates $F^\times$. $\square$

Zorn's lemma is one of the most important results in (axiomatic) set theory, and it's one of the equivalents to the Axiom of Choice. It has a vast array of applications across the fields of pure mathematics, from analysis to (infinite) combinatorics to the area of our class, algebra. Sadly, it's one of the results whose proofs get swept under the rug in pure math classes, and we attempt to go over just enough facts about ordinals (and axiomatic set theory) to be able to prove Zorn's lemma. Anyone interested in more of these (optional) material can check Chapters 6 and 8 of *Introduction to Set Theory* by Hrbacek and Jech.

Let's start off with informal description ("definition") of ordinal numbers. We know that natural numbers start with $0, 1, 2, \dots, n, n+1, \dots$ and continue in this manner. The natural numbers are (the initial segments of) ordinals. But we can continue beyond the natural number by adding ω , the first ordinal number strictly larger than all the natural numbers. Then, we can continue again by adding $\omega + 1, \omega + 2, \dots, \omega + n, \dots$ and so on.

Then, we can add $\omega \times 2$, the first ordinal strictly larger than all the $\omega + n$'s (and of course, all the n 's). Ordinal numbers are informally, the numbers which we can obtain by keep adding 1 (called the "successor step") and then, at the "limit", take the supremum (e.g ω and $\omega \times 2$). Below is an illustration of this written more compactly:

$$0 < \underbrace{1 < 2 < \dots < n < n+1 < \dots}_{\text{Successors}} < \underbrace{\omega}_{\text{Limit}} < \underbrace{\omega+1 < \omega+2 < \dots < \omega+n < \dots}_{\text{Successors}} < \underbrace{\omega \times 2}_{\text{Limit}}$$

We can always add more and more of these ordinals by taking the successor and limits as before, without end. It turns out, there are so many of these ordinal numbers that we can't even put it in a set! That is the statement of the next lemma:

Lemma 3 (Optional Burali-Forti Paradox). The Class ("Collection") of all ordinals is not a set (in set theoretic language, we say the ordinals form a *proper class*).

Proof Outline. Suppose λ is the set of all ordinals. Then, λ itself must be an ordinal (due to technical reasons). But then, $\lambda \in \lambda$ but due to the way ordinals are constructed, $\lambda < \lambda$, a contradiction. $\square$

Let's unpack in more detail the way ordinals are constructed (as described above). It turns out, it's very useful to construct the ordinals α as the set of all ordinal β 's which are strictly less than α . That is, $\alpha = \{\beta \mid \beta < \alpha\}$. Then, as a *set*, $0 = \emptyset$ since there's nothing below 0, $1 = \{0\} = \{\emptyset\}$, $2 = \{0, 1\} = \{\emptyset, \{\emptyset\}\}$, etc, and $\omega = \{n \mid n < \omega\} = \mathbb{N}$.

As a corollary to the Burali-Forti paradox, we obtain Hartog's lemma:

Lemma 4 (Optional Hartog's Lemma). For every set X , there exists an ordinal $\lambda = \{\alpha \mid \alpha < \lambda\}$ which does not inject into X . That is, there is no injection $f : \lambda \hookrightarrow X$.

Proof Outline. The class of ordinals α which *does* inject into X forms a set. Hence, it cannot contain all ordinals. $\square$

We have all the tools to prove Zorn's lemma. But before stating Zorn's lemma, let us recall the following definitions:

Definition 4 (Partial Orders). Let X be a set. We say a binary relation $\preceq$ is a partial order if it satisfies the following conditions:

- (i) (Reflexivity) $x \preceq x$ for all $x \in X$
- (ii) (Anti-Symmetry) Let $x, y \in X$. If $x \preceq y$ and $y \preceq x$, then $x = y$
- (iii) (Transitivity) Let $x, y, z \in X$. If $x \preceq y$ and $y \preceq z$, then $x \preceq z$.

We call the set X together with a chosen partial order a *partially ordered set* (or poset). A chain $\mathcal{C} \subseteq X$ is a subset where $\preceq$ restricted to $\mathcal{C}$ is a linear order: i.e., in addition, it satisfies for all $x, y \in \mathcal{C}$, $x \preceq y$ or $y \preceq x$. An *upper bound* of a subset $\mathcal{C}$ (may or may not be a chain) is any L such that $\forall x \in \mathcal{C}, x \preceq L$. A maximal element $M \in X$ is an element where there is no $x \in X$ with $M \not\preceq x$. Note a poset can have multiple maximal elements, as they need not be greater than every other element.

Axiom 1 (Axiom of Choice). Let $\mathcal{C} = \{C_\alpha\}_{\alpha \in I}$ be a family of non-empty sets ($\forall \alpha \in I, C_\alpha \neq \emptyset$). Then, there exists a *choice function* $f : \mathcal{C} \rightarrow \bigcup C = \bigcup_{\alpha \in I} C_\alpha$ such that $f(C_\alpha) \in C_\alpha$ (i.e. which "chooses" an element $f(C_\alpha)$ from each C_α).

Theorem 2 (Zorn's Lemma). Let $X \neq \emptyset$. If every chain $\mathcal{C} \subseteq X$ has an upper bound $L \in X$, then X has a maximal element.

It turns out, the Axiom of Choice and Zorn's lemma are equivalent under ZF.

Proof that AC $\iff$ Zorn's lemma, Optional.

(Zorn's lemma $\implies$ AC Part:) Take

$$X = \{f : \mathcal{S} \rightarrow \bigcup_{\alpha \in I} C_\alpha \mid \mathcal{S} \subseteq \mathcal{C} \text{ and } f(C_\alpha) \in C_\alpha \text{ for all } C_\alpha \in \mathcal{S}\}$$

In other words, X is the set of all *partial* choice functions. We turn X into a poset by $f \preceq g$ if and only if $\text{dom}(f) \subseteq \text{dom}(g)$ and $g \upharpoonright \text{dom}(f) = f$ (i.e. g extends f).

We check that 1) X satisfies the hypotheses of Zorn's lemma, and thus, X has a maximal element, and 2) a maximal element f is in fact, our desired choice function with full domain $\text{dom}(f) = \mathcal{C}$.

First, it's clear that X is non-empty. Now, let $\mathcal{D} \subseteq X$ be any chain. We'll find a partial choice function to serve as an upper bound of $\mathcal{D}$. Set $\mathcal{S} = \bigcup_{g \in \mathcal{D}} \text{dom}(g)$, and define $L : \mathcal{S} \rightarrow \bigcup_{\alpha \in I} C_\alpha$ by $\forall C_\alpha \in \mathcal{S}, C_\alpha \mapsto g(C_\alpha)$ for any $g \in \mathcal{D}$ such that $C_\alpha \in \text{dom}(g)$. By definition, L is clearly a (partial) choice function, provided it is well-defined (since $g(C_\alpha) \in C_\alpha$ by choice of g). To check well-definedness, fix C_α and consider any two g_1, g_2 with $C_\alpha \in \text{dom}(g_1)$ and $C_\alpha \in \text{dom}(g_2)$. Since $\mathcal{D}$ is a chain, either $g_1 \preceq g_2$ or $g_2 \preceq g_1$ (without loss of generality, assume the former). Then, $\text{dom}(g_1) \subseteq \text{dom}(g_2)$ and $g_2 \upharpoonright \text{dom}(g_1) = g_1$, so $g_2(C_\alpha) = g_1(C_\alpha)$.

So we checked that every chain $\mathcal{D} \subseteq X$ has an upper bound $L \in X$. Hence, by Zorn's lemma, X has a maximal element $f \in X$. We will show that $\text{dom}(f) = \mathcal{C}$. Suppose not; then choose an element $C_\alpha \in \mathcal{C} - \text{dom}(f)$. Then, take $\tilde{f}$ with $\text{dom}(\tilde{f}) = \text{dom}(f) \cup \{C_\alpha\}$ and $\tilde{f} \upharpoonright \text{dom}(f) = f$ and $\tilde{f}(C_\alpha) \in C_\alpha$ (just pick any element of C_α). This clearly defines a strictly larger extension $\tilde{f}$ of f , contradicting maximality of f .

(AC $\implies$ Zorn's lemma Part:) Assume for the sake of contradiction that X satisfies the hypothesis of Zorn's lemma (non-empty, every chain has an upper bound), but has no maximal element. Define a choice function $f : P(X) - \{\emptyset\} \rightarrow X$ with $f(A) \in A$. By Hartog's lemma, choose an ordinal λ which does not inject into X . We will construct by transfinite recursion, an injective map from λ into X .

Define $a_0 = f(X)$ (i.e. we're picking an element of X). Define $a_1, a_2, \dots, a_n, a_{n+1}, \dots$ by $a_1 = f(\{x \mid x \succ a_0\}), a_2 = f(\{x \mid x \succ a_1\}), \dots, a_{n+1} = f(\{x \mid x \succ a_n\})$. In other words, we're just picking $a_0 < a_1 < a_2 < \dots < a_n < a_{n+1} < \dots$. Now, the a_n 's clearly form a chain, so set $a_\omega = f(\{L \mid L \text{ is an upper bound of } \{a_n \mid n < \omega\}\})$. Here's we're using f to pick an upper bound of a_n 's (which exists thanks to the hypothesis of Zorn's lemma). We can continue as in the construction of a_n 's to construct $a_{\omega+n}$'s. Then, again, we have $a_0 < a_1 < \dots < a_n < \dots < a_\omega < a_{\omega+1} < \dots < a_{\omega+n} < \dots$ is a chain, so we can pick an upper bound of it, call it $a_{\omega \times 2}$. We can continue in this way.

More formally, we're doing the following transfinite recursion:

- (i) (Base Case) $a_0 = f(X)$
- (ii) (Successor Case) $a_{\alpha+1} = f(\{x \mid x \succ a_\alpha\})$

- (iii) (Limit Case) Say, β is a limit ordinal (not 0 and not expressible as $\alpha + 1$, e.g. $\omega, \omega \times 2$, etc.). Then, $a_\beta = f(\{L \mid L \text{ is an upper bound of } \{a_\alpha \mid \alpha < \beta\}\})$.

This defines a_α for all $\alpha < \lambda$. Moreover, this is a strictly increasing (and hence, injective) sequence $a : \lambda = \{\alpha \mid \alpha < \lambda\} \hookrightarrow X$, contradicting the fact that λ does not inject into X . $\square$

Remark. While the proof of Zorn's lemma and the preceding lemmas are beyond the scope of this class (and really belong in a set theory course), the *use* of Zorn's lemma is within the scope of this class. Often, you will be asked to use them to prove some theorems in group, ring, or field theory. Furthermore, I believe anyone studying pure mathematics at a deeper level should be comfortable applying Zorn's lemma. Below are a list of theorems from various fields of mathematics which require Zorn's lemma (or its equivalent Axiom of Choice+transfinite recursion) to prove.

Example 6 (Theorems Requiring Zorn's lemma).

- (i) (Functional Analysis, taught in 245B) Hahn-Banach Theorem
- (ii) (Measure Theory, taught in 245A or 245B) There exists a non-Lebesgue measurable subset of $\mathbb{R}$
- (iii) (Ring Theory, taught in 210B) Every proper ideal in a commutative ring (with unit) can be extended to a maximal ideal
- (iv) (Module Theory, taught in 210B?) Baer's Criterion: let R be a (unital) ring. An R -module M is injective if and only if for every left-ideal $I \subseteq R$, every R -linear $f : I \rightarrow M$ can be extended to $f : R \rightarrow M$.
- (v) (General Topology, taught in ??) Alexander Subbasis theorem, used in proof of Tychonoff's theorem (also equivalent to the Axiom of Choice)
- (vi) (Combinatorics/Set Theory, taught in ??) Every (infinite) connected graph has a spanning tree

We finish today's discussion session with an example application of Zorn's lemma in group theory.

Theorem 3. Let $G \neq \{e\}$ be a group. If G is finitely generated, (i.e. $G = \langle g_1, g_2, \dots, g_N \rangle$ for some $g_1, g_2, \dots, g_N \in G$) then G has a maximal proper subgroup.

Proof. Let $X = \{H \mid H \subsetneq G\}$, the set of all proper subgroups of G . Clearly, $\{e_G\} \in X$ (this is where we use G is nontrivial), so $X \neq \emptyset$. A maximal proper subgroup is clearly a maximal element in X , so we will show that X satisfies the hypothesis of Zorn's lemma.

Let $\mathcal{C} \subseteq X$ be a chain of proper subgroups of G . We will find an upper bound $L \in X$, a proper subgroup of G which contains all $H \in \mathcal{C}$. Set $L := \bigcup_{H \in \mathcal{C}} H$. First, we check that L is a subgroup of G to begin with. Let $g, h \in L$. Then, $g \in H_1$ and $h \in H_2$ for some $H_1, H_2 \in \mathcal{C}$. Then, since $\mathcal{C}$ is a chain, $H_1 \leq H_2$ or $H_2 \leq H_1$ (and assume without loss of generality that it's the former). Then, $g, h \in H_2$ so $gh \in H_2$. Also, clearly, $e \in L$ since every $H \in \mathcal{C}$ contains e , and inverses can be checked similarly as well.

Now, we check that L is proper. Suppose $L = G$. Then, $g_1, g_2, \dots, g_N \in L$, so $g_1 \in H_1, g_2 \in H_2, \dots, g_N \in H_N$ for some $H_1, H_2, \dots, H_N \in \mathcal{C}$. But again, since $\mathcal{C}$ forms a chain, $H_{\sigma(1)} \leq H_{\sigma(2)} \leq \dots \leq H_{\sigma(N)}$ for some permutation $\sigma \in S_N$. Then, $g_1, g_2, \dots, g_N \in H_{\sigma(N)}$, so $G = \langle g_1, g_2, \dots, g_N \rangle \leq H_{\sigma(N)}$, contradicting that H_N was proper. $\square$

Discussion 2 - First Isomorphism Theorem, Group Actions

For today's discussion, we will illustrate how the 1st isomorphism theorem can make our HW 1 problem solutions nicer. For the remainder of the time, we will define group action and work on some qual problems related to that.

Definition 5 (Group Homomorphism). A map $\phi : G \rightarrow H$ is a homomorphism if $\phi(gh) = \phi(g)\phi(h)$.

Exercise 5. Let $\phi : G \rightarrow H$ be a homomorphism. Then, $\phi(e_G) = e_H$ and $\phi(g^{-1}) = \phi(g)^{-1}$.

Remark. So, for groups, it's true that every homomorphism sends the identity to identity without additional assumptions. But for structures that aren't groups (e.g. Rings), one may need to assume that the homomorphism sends the identity to the identity.

Theorem 4 (First Isomorphism Theorem). Let $\phi : G \rightarrow H$ be a homomorphism. Then, $\ker \phi \trianglelefteq G$ and $\text{im}(\phi) \leq H$. Moreover, for any $N \trianglelefteq G$ with $N \leq \ker \phi$, there exists a unique $\tilde{\phi} : G/N \rightarrow H$ such that $\tilde{\phi} \circ \pi = \phi$, where $\pi : G \rightarrow G/N$ is the quotient map.

We have the following commutative diagram:

$$\begin{array}{ccc} G & & \\ \downarrow \pi & \searrow \phi & \\ G/N & \xrightarrow{\exists! \tilde{\phi}} & H \end{array}$$

In addition, if $N = \ker \phi$,

then the induced $\tilde{\phi}$ is injective.

HW 1 Problem 9: a) Show that $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ and b) $n\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z}$.

Hints. For part a), consider $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ given by $k \mapsto (k + 2\mathbb{Z}, k + 3\mathbb{Z})$. Check that $\ker = 6\mathbb{Z}$, so it induces an injection $\mathbb{Z}/6\mathbb{Z} \hookrightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. But since they are both finite groups of same cardinality, the induced injection must in fact, be surjective.

For part b), it's easier (in my opinion) to construct the isomorphism $\mathbb{Z}/m\mathbb{Z} \xrightarrow{\cong} n\mathbb{Z}/mn\mathbb{Z}$. Now, consider the maps $\mathbb{Z} \xrightarrow{\cong} n\mathbb{Z} \rightarrow n\mathbb{Z}/mn\mathbb{Z}, k \mapsto nk \mapsto nk + mn\mathbb{Z}$. This is clearly surjective, so it's enough to show that the kernel is $m\mathbb{Z}$, which is left as an exercise. $\square$

HW 1 Problem 2 b): If G is cyclic of order n , and $k \mid n$, then $\exists! H \leq G$ with $|H| = k$.

Hints: Consider $\phi : \mathbb{Z} \rightarrow G = \langle g \rangle$, given by $k \mapsto g^k$. This is a surjective homomorphism with kernel $n\mathbb{Z}$ (exercise!). So, use HW 1 Problem 10 where it says the assignment $H \mapsto \phi^{-1}(H)$ induces a bijection between subgroups of G and subgroups of $\mathbb{Z}$ containing $\ker \phi = n\mathbb{Z}$. So this reduces the problem to finding subgroups of $\mathbb{Z}$ containing $n\mathbb{Z}$, which is much simpler to do. Check that all subgroups of $\mathbb{Z}$ are of the form $d\mathbb{Z}$, and $n\mathbb{Z} \leq d\mathbb{Z}$ if and only if $d \mid n$. The subgroup H of size k , then is just the subgroup corresponding to $\frac{n}{k}\mathbb{Z}$. $\square$

HW 1 Problem 3 b): If G has finitely many subgroups, then G is finite.

Hints. We know that $G = \bigcup_{g \in G} \langle g \rangle$. There must be finitely many $\langle g \rangle$'s by assumption. So the conclusion follows once you show that each $\langle g \rangle$ must be a finite group (exercise!). $\square$

HW 1 Problem 6: Classify G with $|G| \leq 5$.

Hints: Note the cases when $|G| = 1, 2, 3, 5$ is trivial, as G is forced to be cyclic (why?). For the case $|G| = 4$, we can use the following trick. We know $\forall g \in G, \text{ord}(g) = 1, 2$, or 4 . Then, we can divide into two cases:

- (i) $\exists g \in G$ such that $\text{ord}(g) = 4$.

In that case, G is cyclic of order 4.

- (ii) $\forall g \in G, \text{ord}(g) = 1 \text{ or } 2$.

Here, $\forall g \in G, g^2 = e$, so by one of the HW 1 problems, G is abelian. Then, we can either manually show that $G \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (choose $a \neq e$, and $b \notin \{e, a\}$), or do this trick. We can define a $\mathbb{Z}/2\mathbb{Z}$ -scalar multiplication on G by $\mathbb{Z}/2\mathbb{Z} \times G \rightarrow G, (k+2\mathbb{Z}, g) \mapsto g^k$, which is well-defined thanks to the condition that $g^2 = e$. This turns G into a $\mathbb{Z}/2\mathbb{Z}$ -vector space! The dimension of G must be forced to be 2, and G isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

$\square$

HW 1 Problem 7: Find all subgroups of S_3 . Which ones are normal?

The following lemma will be helpful.

Lemma 5. Let $\sigma \in S_n$ and $(a_1, a_2, \dots, a_k)$ be a k -cycle in S_n . Then, $\phi := \sigma \circ (a_1, a_2, \dots, a_k) \circ \sigma^{-1} = (\sigma(a_1), \sigma(a_2), \dots, \sigma(a_k))$.

Proof. We'll check what ϕ does to $\sigma(a_1), \dots, \sigma(a_k)$. We see that

$$\begin{array}{ccc} \sigma(a_1) & \mapsto & a_1 \mapsto a_2 \mapsto \sigma(a_2) \\ \sigma(a_2) & \mapsto & a_2 \mapsto a_3 \mapsto \sigma(a_3) \\ & \vdots & \\ \sigma(a_k) & \mapsto & a_k \mapsto a_1 \mapsto \sigma(a_1) \end{array}$$

Now, for $b \notin \{\sigma(a_1), \dots, \sigma(a_k)\}$, we see that $\sigma^{-1}(b) \notin \{a_1, \dots, a_k\}$. So the map ϕ sends $b \mapsto \sigma^{-1}(b) \mapsto \sigma^{-1}(b) \mapsto b$, which is itself. Hence, ϕ is in fact, the k -cycle $(\sigma(a_1), \sigma(a_2), \dots, \sigma(a_k))$. $\square$

Corollary 1. Two permutations in S_n are conjugate if and only if they have the same cycle type in their decomposition into disjoint cycles.

For the remainder of today's session, we will go over group actions. First, recall the following definition.

Definition 6 (Group Action). A (left) group action $G \curvearrowright X$ is a map $\cdot : G \times X \rightarrow X$ satisfying the following properties:

- (i) $e \cdot x = x$
- (ii) $(gh) \cdot x = g \cdot (h \cdot x)$.

Since many groups come from automorphisms of a set X with additional structures, $\text{Aut}(X) \curvearrowright X$ via compositions.

Example 7 (Group Action Examples).

- (i) $S_X \curvearrowright X$ by $\sigma \cdot x = \sigma(x)$.
- (ii) $\text{Gl}(V) \curvearrowright \{(v_1, v_2, \dots, v_k) \mid v_1, v_2, \dots, v_k \text{ are linearly independent}\}$ via $T \cdot (v_1, v_2, \dots, v_k) = (Tv_1, Tv_2, \dots, Tv_k)$.
- (iii) $D_{2n} \curvearrowright \{\text{Vertices of } n\text{-gons}\}$
- (iv) $\text{Aut}_{\mathbb{R}}(\mathbb{C}) \curvearrowright \{\pm i\}$ via $\sigma \cdot \pm i = \pm \sigma(i)$.

We also have more abstract examples.

Example 8.

- (i) (Left Action) $G \curvearrowright G$ by $g \cdot k = gk$. More generally, if $H \leq G$, then $H \curvearrowright G$ by $h \cdot k = hk$. Note the orbits (to be defined later) $H \cdot k$ of the action are precisely the right cosets Hk of G .
- (ii) (Action by Conjugation) $G \curvearrowright G$ by $g \cdot k = gkg^{-1}$. Similarly, have $G \curvearrowright \{H \mid H \leq G\}$ via $g \cdot H = gHg^{-1}$.

Theorem 5 (Correspondence Between Group actions and Homomorphism into S_X). Let $G \curvearrowright X$. Then, we can induce a homomorphism $\alpha : G \rightarrow S_X$ via $\alpha(g) := \sigma_g : (x \mapsto g \cdot x)$. Conversely, if $\alpha : G \rightarrow S_X$ is a homomorphism, then we can induce the action $G \curvearrowright X$ by $g \cdot x = \alpha(g)(x)$.

Moreover, the two assignments $\cdot \mapsto \alpha$ and $\alpha \mapsto \cdot_\alpha$ give rise to bijections between the set of groups actions $G \curvearrowright X$ and homomorphisms $G \rightarrow S_X$ which are inverses of one another.

Proof. We will check that $\alpha(gh)(x) = (\alpha(g) \circ \alpha(h))(x)$ for all $g, h \in G$ and $x \in X$, so that α is a homomorphism as long as $\alpha(g) \in S_X$ for all $g \in G$. Check

$$\begin{aligned} \alpha(gh)(x) &= (gh) \cdot x \\ &= g \cdot (h \cdot x) \\ &= \alpha(g)(\alpha(h)(x)) \\ &= (\alpha(g) \circ \alpha(h))(x) \end{aligned}$$

Now, we check $\alpha(g) \in S_X$. Note $\alpha(e) : x \mapsto e \cdot x = x$, so $\alpha(e) = \text{id}_X$. So $\text{id}_X = \alpha(e) = \alpha(g \circ g^{-1}) = \alpha(g^{-1} \circ g) = \alpha(g) \circ \alpha(g^{-1}) = \alpha(g^{-1}) \circ \alpha(g)$. Hence, $\alpha(g)$ has an inverse given by $\alpha(g^{-1})$, so $\alpha(g)$ is in fact a bijection.

Next, we will check given $\alpha : G \rightarrow S_X$, the map given by $g \cdot x = \alpha(g)(x)$ is a valid group action. Again, check $e \cdot x = \alpha(e)(x) = \text{id}_X(x) = x$, and $(gh) \cdot x = \alpha(gh)(x) = (\alpha(g) \circ \alpha(h))(x) = \alpha(g)(\alpha(h)(x)) = g \cdot (h \cdot x)$.

Finally, we check that $\cdot \mapsto \alpha$ and $\alpha \mapsto \cdot_\alpha$ are inverses of one another. Consider $\cdot \mapsto \alpha \mapsto \cdot_\alpha$. We see that $g \cdot_\alpha x = \alpha(g)(x) = g \cdot x$. Similarly, consider $\alpha \mapsto \cdot \mapsto \alpha$. We see that $\alpha(g) : x \mapsto g \cdot x = \alpha(g)(x)$. So α and α agrees. $\square$

Corollary 2 (Cayley's Theorem). Every group G embeds into S_G .

Proof. The left action $G \curvearrowright G, g \cdot k = gk$ induces a homomorphism $\alpha : G \rightarrow S_G$. We just need to check that $\ker \alpha$ is trivial. Note $g \in \ker \alpha$ if and only if $g \cdot k = k$ for all $k \in G$, but this can only happen if $g = e$. $\square$

We will end the session with one qual problem related to Cayley's embedding theorem.

Spring 2019 Problem 8: Prove that every finite group of order n is isomorphic to a subgroup $\text{Gl}_{n-1}(\mathbb{C})$.

Discussion 3 - Group Action Qual Problems

We'll finish off the problem from last week's discussion which is given above. Note every G embeds into $S_G \cong S_n$, so it's enough to find an embedding of S_n into $\text{Gl}_{n-1}(\mathbb{C})$, which is stated as a lemma below:

Lemma 6. Every S_n embeds into $\text{Gl}_{n-1}(\mathbb{C})$.

Proof. For $n = 2$, S_2 clearly embeds into $\text{Gl}_1(\mathbb{C}) = \mathbb{C}^\times$, so assume $n > 2$. First, embed S_n into $\text{Gl}_n(\mathbb{C})$ via the group action $S_n \curvearrowright \mathbb{C}^n$ with $\sigma \cdot e_i = e_{\sigma(i)}$. This induces an embedding $\sigma \mapsto P_\sigma$ where P_σ is the linear map taking e_i to $e_{\sigma(i)}$.

Note now, that $e_1 + e_2 + \cdots + e_n$ is an eigenvector of all P_σ 's (as $\sigma \in S_n$) of eigenvalue 1. Hence, if we take the matrix of P_σ with respect to the basis, say $\beta = (e_1 + e_2 + \cdots + e_n, e_2, e_3, \dots, e_n)$, then

$$[P_\sigma]_\beta = \begin{pmatrix} 1 & * \\ 0 & A_\sigma \end{pmatrix}$$

for $A_\sigma \in \text{Gl}_{n-1}(\mathbb{C})$.

Claim. The assignment $\sigma \mapsto A_\sigma$ is a homomorphism from S_n to $\text{Gl}_{n-1}(\mathbb{C})$.

Proof of the Claim. Take $\tau, \sigma \in S_n$. Then,

$$\begin{aligned} [P_{\tau \circ \sigma}]_\beta &= [P_\tau]_\beta [P_\sigma]_\beta \\ &= \begin{pmatrix} 1 & * \\ 0 & A_\tau \end{pmatrix} \begin{pmatrix} 1 & * \\ 0 & A_\sigma \end{pmatrix} \\ &= \begin{pmatrix} 1 & * \\ 0 & A_\tau A_\sigma \end{pmatrix} \end{aligned}$$

$\square$

We finally check that the homomorphism $S_n \rightarrow \text{Gl}_{n-1}(\mathbb{C})$ has a trivial kernel, so that it is indeed an embedding. Suppose $A_\sigma = \text{Id}_{n-1}$. Then,

$$[P_\sigma]_\beta = \begin{pmatrix} 1 & * \\ 0 & \text{Id}_{n-1} \end{pmatrix}$$

In particular, for each $j \geq 2$, $\sigma(e_j) = e_{\sigma(j)} = k_j(e_1 + e_2 + \cdots + e_n) + e_j$. Since all but 1 of the coefficients of $e_1, e_2, \dots, e_n$ are 0, k_j is forced to be 0 (this is where we use $n > 2$; otherwise, $k_j = -1$ works just as well). Hence, P_σ is the identity, and σ is forced to be the identity as well. $\square$

For today's session, we will go over three qual problems involving group actions in some way. First, recall the definition.

Definition 7 (Group Action). We say G acts on X (on the left) and denote $G \curvearrowright X$ if

- (i) $e \cdot x = x$
- (ii) $(gh) \cdot x = g \cdot (h \cdot x)$

In preparation for the problems, we will recall various basic definitions and facts about group actions.

Definitions and Facts 1. Let $G \curvearrowright X$.

- (i) There is a one-to-one correspondence between group actions $G \curvearrowright X$ and homomorphisms $G \rightarrow S_X$.
- (ii) $\text{Orbit}(x) = G \cdot x = \{g \cdot x \mid g \in G\}$
- (iii) $\text{Stab}(x) = G_x = \{g \in G \mid g \cdot x = x\}$
- (iv) If we define $x \sim y$ by $\exists g \in G, y = g \cdot x$, this turns $\sim$ into an equivalence relation with $[x]_\sim = G \cdot x$. In particular, orbits (hence the equivalence classes) partition X .
- (v) (Orbit Stabilizer Theorem) There is a bijection $f : G/G_x \rightarrow G \cdot x$ given by $f(gG_x) = g \cdot x$. Moreover, if $G \curvearrowright G/G_x$ acts by the left multiplication, then f is G -equivariant, i.e. $f(k \cdot gG_x) = k \cdot (g \cdot x)$.
- (vi) We say $G \curvearrowright X$ is transitive if $G \cdot x = X$ for some (and hence, all) $x \in X$.

Using group actions on the left, we can solve one of the homework problems very cleanly.

HW 2 Problem 6: If $H \leq G$ and $[G : H] = n < \infty$, then there exists a normal subgroup $N \trianglelefteq G$ contained in H ($N \leq H$) such that $[G : N] \mid n!$.

Hints: Consider $G \curvearrowright G/H$ by left multiplication $g \cdot kH = gkH$. This induces a homomorphism $\alpha : G \rightarrow S_{G/H} \cong S_n$. Check that $\ker \alpha$ is our desired normal subgroup. $\square$

Remark. This is (to my knowledge) called Cayley's theorem (on normal subgroups).

The following statement is tested in various recent years, and I believed it was important enough to reproduce the statement and the proof even if covered in lecture.

Qual Problem from Various Years: If p is the smallest prime divisor of $|G|$, then every $H \leq G$ of index $[G : H] = p$ is normal.

Proof. Choose $N \leq H$ which is normal in G and such that $[G : N] \mid p!$. Note, then $[H : N] = [G : N]/[G : H]$ divides $p!/p = (p-1)!$. But $[H : N] \mid |G|$ as well, so $[H : N] \mid \gcd(|G|, (p-1)!) = 1$, hence $[H : N] = 1$. In particular, $H = N$ and H is normal since N is chosen to be normal. $\square$

Fall 2023 Problem 1: Let G be a group, let $H \subset G$ be a subgroup of finite index $n \geq 2$, and let $x \in G$. Prove that $[H : H \cap xHx^{-1}] \leq n - 1$.

Proof. Consider $H \curvearrowright G/H$ on the left by $h \cdot gH = hgH$. Note H is a fixed point of the action since $\forall h \in H, h \cdot H = hH = H$. Then, since $\{H\}$ is an orbit of a size $1 < n$, the action cannot be transitive. Therefore, for all $x \in G$, $\text{Orbit}(xH)$ has size at most $n - 1$. But by the orbit-stabilizer theorem, $H/\text{Stab}(xH)$ bijects onto $\text{Orbit}(xH)$, so $[H : \text{Stab}(xH)] \leq n - 1$.

It turns out, $\text{Stab}(xH) = H \cap xHx^{-1}$, and it will finish the proof. For that, for any $h \in H$,

$$\begin{aligned} h \cdot xH = xH &\iff hxH = xH \\ &\iff x^{-1}hxH = H \\ &\iff x^{-1}hx \in H \\ &\iff h \in xHx^{-1} \end{aligned}$$

This in turn, proves that $h \in \text{Stab}(xH)$ if and only if $h \in H \cap xHx^{-1}$. $\square$

In preparation for the next problem, we will define the quaternion group.

Definition 8 (Quaternion Group). The *Quaternion group* is a set $\{\pm 1, \pm i, \pm j, \pm k\}$ together with the unique operation satisfying $(\pm 1)^2 = 1, a^2 = -1$ for $a \in \{i, j, k\}$, $(-1) \cdot a = -a$ for $a \in \{i, j, k\}$, and $ij = k, ji = -k$ (and from there, it follows that $jk = i, kj = -i, ki = j, ik = -j$).

Exercise 6. Check that the operation defined above indeed does turn the set $\{\pm 1, \pm i, \pm j, \pm k\}$ into a group.

Fall 2018 Problem 1: Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ be the quaternion group of order 8.

(1) Show that every non-trivial subgroup of Q_8 contains -1 .

(2) Show that Q_8 does not embed in the symmetric group S_7 (as a subgroup).

Proof of (1): Suppose H is a nontrivial subgroup of Q_8 . Then, choose an element $g \in H$ not equal to 1. If $g = -1$, we're done; otherwise, $g = \pm a$ where $a \in \{i, j, k\}$, so $g^2 = (\pm a)^2 = -1 \in H$. $\square$

Proof of (2): We'll show that no homomorphism $\alpha : Q_8 \rightarrow S_7$ is injective. Recall that a homomorphism $\alpha : Q_8 \rightarrow S_7$ induces a group action $Q_8 \curvearrowright [7]$. Now, note for all $k \in [7]$, $Q_8 \cdot k \subseteq [7]$, so in particular, cannot have size 8. Hence, by orbit-stabilizer theorem, $Q/(Q_8)_k$ also cannot be of size 8, so $(Q_8)_k$ is non-trivial. By part (1), $-1 \in (Q_8)_k$. So we have for all $k \in [7]$, $(-1) \cdot k = \alpha(-1)(k) = k$, so $\alpha(-1) = \text{id}_{[7]}$. Hence, $-1 \in \ker \alpha$ so α cannot be an embedding. $\square$

Fall 2018 Problem 2: Let G be a finitely generated group having a subgroup of finite index $n > 1$. Show that G has finitely many subgroups of index n and has a proper characteristic subgroup (i.e preserved by all automorphisms) of finite index.

We are proving two things: (1) there are only finitely many subgroups of index n , and (2) (using (1)) there is a proper characteristic normal subgroup of finite index. The idea behind the first proof is as follows: we notice if we have a subgroup H of finite index n , then $G \curvearrowright G/H$ on the left transitively, and $H = G_H$. So, it's enough to show that there are only finitely many group actions $G \curvearrowright [n]$, as we will formalize below:

Proof of (1). First, we claim that there are only finitely many group actions $G \curvearrowright [n]$. To see this, take a finite set of generators $g_1, g_2, \dots, g_N$. Then, $G \curvearrowright [n]$ is uniquely determined by $g_j \cdot k$ where $j = 1, 2, \dots, N$ and $k \in [n]$, so there are only finitely many such group actions (note a given choice of $g_j \cdot k$ may not induce a group action, but if two actions agree on $g_j \cdot k$, then they must be equal).

Now, consider the map

$$f : \{\cdot \mid \cdot \text{ is a group action } G \curvearrowright [n]\} \rightarrow \{H \mid H \leq G, [G : H] \leq n\}$$

defined by $G \curvearrowright X \mapsto G_1$. Given every $H \leq G$ of index n , we have a left group action $G \curvearrowright G/H$. After choosing a bijection $\phi : G/H \rightarrow [n]$ with $\phi(H) = 1$, we may induce the unique group action $G \curvearrowright [n]$ so that $\phi : G/H \rightarrow [n]$ is G -equivariant. Then, clearly, $H = G_H = G_1$, so H lies in the image of f . So, we have $\{H \mid H \leq G, [G : H] = n\} \subset \text{Im}(f)$. But since the domain of f , the set of all group actions $G \curvearrowright [n]$ is finite, so is the $\text{Im}(f)$, and hence the subset $\{H \mid H \leq G, [G : H] = n\}$. $\square$

Proof of (2). Let $H_1, H_2, \dots, H_k$ be the set of all index n subgroups of G . We will set $N = H_1 \cap H_2 \cap \dots \cap H_k$ to be our desired characteristic normal subgroup. First, N is clearly proper since H_1 is. Next, let's show N is characteristic: let $\phi \in \text{Aut}(G)$, then $\phi(N) = \phi(H_1) \cap \phi(H_2) \cap \dots \cap \phi(H_k)$. But since $H \mapsto \phi(H)$ is a bijection from $\{H_1, H_2, \dots, H_k\} = \{H \mid H \leq G, [G : H] = n\}$ to itself, $\phi(H_1) \cap \phi(H_2) \cap \dots \cap \phi(H_k) = H_1 \cap H_2 \cap \dots \cap H_k = N$. This proves the claim that N is characteristic normal.

Finally, we check that N is of finite index. For this, we use the following definition and lemma:

Definition 9. Let $H \leq G$ and $S \subset G$ containing H . Then, we define S/H to be $\pi(S)$ where $\pi : G \rightarrow G/H$ is the quotient map to the set of cosets. Explicitly, $S/H = \pi(S) = \{gH \mid g \in S\}$.

Lemma 7. Let $H, K \leq G$. Then, $f : H/H \cap K \rightarrow HK/K, hH \cap K \mapsto hK$ is a bijection.

Proof of the Lemma. Let's prove f is well-defined and injective simultaneously. Let $h_1, h_2 \in H$. We know

$$\begin{aligned} h_1H \cap K = h_2H \cap K &\iff h_1^{-1}h_2 \in H \cap K \\ &\iff h_1^{-1}h_2 \in H \text{ and } h_1^{-1}h_2 \in K \\ &\iff h_1K = h_2K \end{aligned}$$

This proves both f is well-defined (independent of choice of $h_2 \in h_1H \cap K$) and injective. For surjectivity, take any aK in the codomain with $a \in HK$. Then, $a = hk$ for some $h \in H$ and $k \in K$, so $aK = hkK = hK = f(hH \cap K)$. $\square$

Now,

$$\begin{aligned} [G : N] &= [G : H_1 \cap H_2 \cap \dots \cap H_k] \\ &= \left| \frac{G}{H_1} \right| \left| \frac{H_1}{H_1 \cap H_2} \right| \dots \left| \frac{H_1 \cap \dots \cap H_{k-1}}{H_1 \cap \dots \cap H_{k-1} \cap H_k} \right| \end{aligned}$$

Using the previous lemma, we see that

$$\left| \frac{H_1 \cap \dots \cap H_i}{(H_1 \cap \dots \cap H_i) \cap H_{i+1}} \right| = \left| \frac{(H_1 \cap \dots \cap H_i)H_{i+1}}{H_{i+1}} \right| \leq \left| \frac{G}{H_{i+1}} \right| = n$$

for all $i = 1, \dots, k-1$. Hence, $[G : N]$ is at most n^k . $\square$

Discussion 4 - Sylow Group Qual Problems

For today's discussion, we will go over qual problems involving Sylow p -subgroups. Recall the following important lemma:

Lemma 8 (Class Equation). Let P be a p -group and $P \curvearrowright X$. Then,

$$|X| = \sum_{|\text{Orbit}| > 1} |\text{Orbit}| + |\text{Fix}(P)|.$$

That is, $|X|$ can be broken down into a sum of sizes of orbits larger than 1 and the sum of sizes of orbits of size exactly 1, the latter being exactly the number of fixed points. In particular, $|X| \equiv |\text{Fix}(P)| \pmod{p}$ since each $|P \cdot x|$ is of prime power.

Next, we will recall several of the important Sylow's theorems together with a summary of their proof techniques.

Theorem 6 (Cauchy's Theorem). Let G be a group with $p \mid |G|$ be a prime. Then, there is an element $a \in G$ with $\text{ord}(a) = p$.

Proof Outline: Take $\mathbb{Z}/p\mathbb{Z} \curvearrowright X$ where $X = \{(a_0, a_1, \dots, a_{p-1}) \in G^p \mid a_0 a_1 \cdots a_{p-1} = e\}$, and the action is given by $\bar{1} \cdot (a_0, a_1, \dots, a_{p-1}) = (a_{p-1}, a_0, \dots, a_{p-2})$. Apply class equation to show $|\text{Fix}(\mathbb{Z}/p\mathbb{Z})| \equiv 0 \pmod{p}$ and the fact that $(e, e, \dots, e) \in \text{Fix}(\mathbb{Z}/p\mathbb{Z})$ to show that in fact, $|\text{Fix}(\mathbb{Z}/p\mathbb{Z})| \geq p$. $\square$

Theorem 7 (First Sylow's Theorem). Let G be a group and $|G| = p^k m$ where $p \nmid m$ (p is a prime). Then, G has a Sylow p -subgroup $P \leq G$ (Sylow p -subgroups are subgroups of order p^k). Moreover, any p -subgroup H can be extended to a Sylow p -subgroup $P \geq H$.

Proof. Apply Cauchy's theorem to find a subgroup of order exactly $p = p^1$. For each p -subgroup $H \leq G$ of order p^i , we can extend it to a p -subgroup $H' \geq H$ of order p^{i+1} as follows: first, have $H \curvearrowright G/H$ on the left. Show $\text{Fix}(H) = N_G(H)/H$, i.e. the fixed points gH are exactly the points gH with $g \in N_G(H)$. As before, apply class equation to get $|N_G(H)/H| = |\text{Fix}(H)| \equiv 0 \pmod{p}$. But $N_G(H)/H$ is a group, so apply Cauchy's theorem on $N_G(H)/H$ and then, pull back via $N_G(H) \twoheadrightarrow N_G(H)/H$. $\square$

Theorem 8 (Second Sylow's Theorem). Any two Sylow p -subgroups of G are conjugate.

Proof Outline: Let P, Q be two Sylow p -subgroups. Take $P \curvearrowright G/Q$. Apply class equation to get $\text{Fix}(P) \neq \emptyset$, and in fact, a fixed point gQ satisfies $P \leq gQg^{-1}$. $\square$

Corollary 3. A Sylow p -subgroup P of G is normal if and only if P is the only Sylow p -subgroup. Equivalently, if we denote n_p the number of Sylow p -subgroups, then P is normal if and only if $n_p = 1$.

Theorem 9 (Third Sylow's Theorem). Let $P \leq G$ be any Sylow p -subgroup. Then, $[G : N_G(P)] = n_p \equiv 1 \pmod{p}$.

Proof Outline: Take $G \curvearrowright X$ where $X = \{P \leq G \mid P \text{ is a Sylow } p\text{-subgroup}\}$ via conjugation. Then, by the second Sylow's theorem, $X = G \cdot P$, so by orbit-stabilizer theorem, $|G \cdot P| = [G : G_P] = [G : N_G(P)]$. For the second equality, restrict the action to $P \curvearrowright X$. Then, we'll see that this now has a unique fixed point P (Q is a fixed point iff $Q \leq N_G(P)$, but we know $P \trianglelefteq N_G(P)$, so $P = Q$ must be the unique normal Sylow p -subgroup of $N_G(P)$). Apply class equation. $\square$

With the techniques for the proofs of each Sylow's theorem in mind, we can quickly solve some of the Sylow group qual problems.

Spring 2022 Problem 3: Let G be a finite p -group and let $H < G$ be a proper subgroup. We write as usual $H^g = gHg^{-1}$ for every $g \in G$.

- (a) Show that the normalizer $N_G(H)$ of H in G is strictly larger than H .
- (b) Show that if H is not normal in G then there exists another proper subgroup $H < K < G$ and $g \in G$ such that $K^g = K$ but $H^g \neq H$.

Proof of (a): As in the proof of First Sylow's theorem, take $H \curvearrowright G/H$ on the left. Then, fixed points are precisely $N_G(H)/H$, which by class equation, has size $0 \pmod{p}$. $\square$

Proof of (b): Take $K = N_G(H) \not\geq H$ by part (a). Note by assumption, K must be a proper subgroup ($N_G(H) = G$ is equivalent to $H \trianglelefteq G$). So again, by part (a), $K \not\leq N_G(K)$. So take $g \in N_G(K) - K$, and we see that the two conclusions are satisfied. $\square$

Fall 2020 Problem 1: Let $p < q < r$ be primes and G a group of order pqr . Prove that G is not simple and, in fact, has a normal Sylow r -group.

Proof. We will first, show that at least one of the p, q, r -Sylow subgroups are normal in G . Suppose otherwise; then,

$$\begin{aligned} n_r &\mid pq \text{ and } n_r \equiv 1 \pmod{r} \\ n_p &\mid qr \text{ and } n_p \equiv 1 \pmod{p} \\ n_q &\mid pr \text{ and } n_q \equiv 1 \pmod{q} \end{aligned}$$

The fact that n_p, n_q, n_r are not all 1 implies that $n_r = pq$, $n_p \geq q$ and $n_q \geq p$ (in fact $n_q \geq r$ but we don't need this).

Choose $R_1, R_2, \dots, R_{pq}$ distinct r -Sylow subgroups, $P_1, P_2, \dots, P_q$ distinct p -Sylow subgroups, and $Q_1, Q_2, \dots, Q_p$ q -Sylow subgroups. Then, for all

$$H \neq K \in \{R_j, P_i, Q_l\}_{j \in [pq], i \in [q], l \in [p]},$$

$H \cap K = \{e\}$. To see this, if H and K are same prime Sylow subgroups, then $|H \cap K| = p^i, q^i$, or r^i for some $i < 1$. The only way this can happen is if $i = 0$, and hence, $|H \cap K| = 1$. If H and K are different prime Sylow subgroups, then $|H \cap K| \mid 1$, so $|H \cap K| = 1$.

Set

$$A = \bigsqcup_{j=1}^{pq} (R_j - \{e\}) \sqcup \bigsqcup_{i=1}^q (P_i - \{e\}) \sqcup \bigsqcup_{l=1}^p (Q_l - \{e\})$$

Then, $|A| = (r-1)pq + (p-1)q + (q-1)p = pqr + pq - p - q > pqr = |G|$, but this contradicts that $A \subseteq G$.

Now, we show that the r -Sylow group must be normal. Take any $H \trianglelefteq G$ a Sylow subgroup (which we showed exists by the previous part). If H is r -Sylow, we're done, so assume H is either p or q Sylow. Then, the quotient G/H is a group of order pr or qr . In either case, the r -Sylow subgroup $\bar{R}$ is normal in G/H . Pulling back via the quotient map $\pi : G \twoheadrightarrow G/H$, we get $\pi^{-1}(\bar{R}) \trianglelefteq G$ is a subgroup of order either pr or qr (where $|H|$ is either p or q , respectively). We may now choose $R \trianglelefteq \pi^{-1}(\bar{R}) \trianglelefteq G$ the unique normal

r -Sylow subgroup of $\pi^{-1}(\overline{R})$. We do not get $R \trianglelefteq G$ immediately yet, but we show $R \trianglelefteq G$ as follows.

Take $g \in G$. Then, $gRg^{-1} \leq g\pi^{-1}(\overline{R})g^{-1} = \pi^{-1}(\overline{R}) \leq G$. The equality follows from the fact that $\pi^{-1}(\overline{R})$ is normal in G . Hence, both R and gRg^{-1} are the unique Sylow r -subgroups of $\pi^{-1}(\overline{R})$, and therefore, $R = gRg^{-1}$. This shows $R \trianglelefteq G$. $\square$

Remark. In general, $K \trianglelefteq N \trianglelefteq G$ does not imply $K \trianglelefteq G$. So we could not conclude immediately from $R \trianglelefteq \pi^{-1}(\overline{R}) \trianglelefteq G$ that $R \trianglelefteq G$. However, if $K \trianglelefteq_{\text{char}} N \trianglelefteq G$ (i.e. if K is characteristically normal in N , and N is normal in G), then it does follow $K \trianglelefteq G$. The proof is very similar to what we did and is left up to you as an exercise.

Spring 2024 Problem 6: Let G be a finite group of order 300. Show that G is not simple by considering its action on its Sylow 5-subgroups.

Proof. As suggested in the problem, we look at the Sylow 5-subgroups. Note $300 = 2^2 \cdot 3 \cdot 5^2$, so $n_5 \mid 2^2 \cdot 3 = 12$ and $n_5 \equiv 1 \pmod{5}$. The only possibilities for n_5 are then, 1 and 6. If $n_5 = 1$, then the unique 5-Sylow subgroup is normal, and we're done. So assume that $n_5 = 6$.

Let $X = \{P_5 \leq G \mid P_5 \text{ is } 5\text{-Sylow}\}$, and let $G \curvearrowright X$ by conjugation. Then, this induces the homomorphism $\alpha : G \rightarrow S_X \cong S_6$, and hence, $N := \ker \alpha \trianglelefteq G$ is a normal subgroup of index $[G : N] \mid 6! = 720$. Since $300 \nmid 6! = 720$, N cannot be the identity subgroup. Also, clearly, $N \leq N_G(P_5)$ for any $P_5 \in X$, so it is clearly not equal to G . $\square$

Remark. Alternatively, the last part can be proven as follows. We know $[G : N_G(P_5)] = 6$, so by applying Cayley's theorem, there exists a normal subgroup $N \trianglelefteq G$ such that $N \leq N_G(P_5)$, and $[G : N] \mid 6!$. Since $300 \nmid 6! = 720$, $N \neq \{e\}$. In fact, check that the two normal subgroups constructed are exactly the same subgroups.

Discussion 5 - Permutation Groups

For today's discussion, we will go over the properties of permutation groups. Let's recall some basic facts:

Definitions and Facts 2.

- (I) (Permutation Group) The permutation group is the set $S_n := \{\sigma[n] \rightarrow [n] \mid \sigma \text{ is a bijection}\}$ together with composition operation.
- (II) Every permutation $\sigma \in S_n$ has a decomposition into product of disjoint cycles. Moreover, such a decomposition is unique up to rearrangement. The cycle type is the lengths of each of the cycles when the cycles are arranged in the order of decreasing lengths. For instance, $(12)(35) \in S_5$ has cycle type 2, 2, 1.
- (III) A cycle can be decomposed further into product of 2-cycles as follows: $(a_1, a_2, \dots, a_k) = (a_1, a_2)(a_2, a_3) \cdots (a_{k-1}, a_k)$. Hence, it follows every permutation $\sigma \in S_n$ can be written as a product of 2-cycles. In other words, the 2-cycles generate S_n .
- (IV) For $\sigma \in S_n$ and a k -cycle $(a_1, a_2, \dots, a_k)$, the following conjugation fact holds: $\sigma(a_1, a_2, \dots, a_k)\sigma^{-1} = (\sigma(a_1), \sigma(a_2), \dots, \sigma(a_k))$. In particular, two permutations are conjugate if and only if they have the same cycle type.

- (V) (Sign of a Permutation) Given a permutation $\sigma \in S_n$, its sign $\text{sgn}(\sigma)$ is equal to $(-1)^k$ where k is the number of 2-cycles in the 2-cycle decomposition of σ . This k is independent modulo 2 of the 2-cycle decomposition and is well defined (more on this later).
- (VI) For all $\sigma, \tau \in S_n$, $\text{sgn}(\tau \circ \sigma) = \text{sgn}(\tau)\text{sgn}(\sigma)$. Hence, the assignment $\text{sgn} : S_n \rightarrow \{\pm 1\}, \sigma \mapsto \text{sgn}(\sigma)$ is a surjective homomorphism (for $n \geq 2$).
- (VII) The kernel of the sign homomorphism is defined as $A_n := \ker \text{sgn}$. In particular, $A_n \leq S_n$ and $|S_n/A_n| = |\{\pm 1\}| = 2$. This group A_n is also called the alternating group.
- (VIII) For $n \geq 5$, the alternating group A_n is simple (and nonabelian). This is the main reason why polynomial equations $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$ (with $a_n \neq 0$) of degree $n \geq 5$ do not admit general formula in terms of radicals. More details to be taught in 210B.

Armed with these facts, we can tackle many of the HW 4 problems.

Homework 4 Problem 5: Show A_4 has no subgroup of order 6.

Hints: Suppose we have $H \leq A_4$ of order 6. Then, H has index 2 so is normal in A_4 . Consider the surjective homomorphism $\pi : A_4 \twoheadrightarrow A_4/H$. Then, for every 3-cycle (a, b, c) , $\pi((a, b, c))$ has order dividing both $3 = \text{ord}((a, b, c))$ and $2 = |A_4/H|$, and hence is the identity. That shows every 3-cycle (a, b, c) belongs to H . $\square$

Homework 4 Problem 6:

- (a) Show that S_n is generated by $(12), (13), \dots, (1n)$.
 (b) Show that S_n is generated by $(12), (12 \cdots n)$.

Hints for (a): We know S_n is generated by the two cycles (ij) . We already have cycles of the form $(1i)$, so it's enough to find cycles of the form (ij) where $i, j \neq 1$ (and $i \neq j$). For this, just use conjugation: $(1j)(1i)(1j)^{-1} = (ji) = (ij)$. $\square$

Hints for (b): Note that $(12)(12 \cdots n) = (23 \cdots n) := \sigma$. In particular, σ fixes 1. Then, by conjugating with $\sigma^?$ (some power of σ), we see that $\sigma^?(12)\sigma^{-?} = (1??)$. We leave up to you to find out what goes into ? and ??. Hence, we can obtain all 2-cycles of the form $(1k)$ and use part (a) to finish the problem. $\square$

Homework 4 Problem 7 Part (a): Show that the centralizer of A_n in S_n is trivial for $n \geq 4$.

Hints: We're showing $C_{S_n}(A_n) = \{\sigma \in S_n \mid \forall \tau \in A_n, \sigma\tau\sigma^{-1} = \tau\} = \{\text{id}\}$. Let $\sigma \in C_{S_n}(A_n)$. Note since all 3-cycles are in A_n , $\sigma \circ (a \ b \ c)\sigma^{-1} = (\sigma(a) \ \sigma(b) \ \sigma(c)) = (a \ b \ c)$. In particular, $\sigma(\{a, b, c\}) = \{a, b, c\}$ for all 3-element subset $\{a, b, c\}$. Specializing further, we see that for all $k \geq 3$, $\sigma(\{1, 2, k\}) = \{1, 2, k\}$. Now, (and this is where we use $n \geq 4$), $\sigma(\{1, 2, 3\} \cap \{1, 2, 4\}) = \{1, 2, 3\} \cap \{1, 2, 4\}$ since σ is injective, so this gives us $\sigma(\{1, 2\}) = \{1, 2\}$. But then, $\forall k \geq 3$, $\sigma(k) = k$ (why?). We can similarly (or by hand) show that $\sigma(1) = 1$ and $\sigma(2) = 2$, whose details are left to you. $\square$

Homework 4 Problem 8: Prove that $\text{Aut}(S_3) = \text{Inn}(S_3)$ and the latter is isomorphic to S_3 via the map $S_3 \rightarrow \text{Inn}(S_3), \sigma \mapsto (\tau \mapsto \sigma\tau\sigma^{-1})$.

Hints: The first part is the harder part, and therefore, the second, more easier part is left to you. Consider an automorphism $\phi \in \text{Aut}(S_3)$. We know that $(1\ 2)$ and $(2\ 3)$ generates S_3 , so it suffices to show that ϕ acts as inner automorphism on these two elements. Moreover, clearly, $\phi((1\ 2))$ and $\phi((2\ 3))$ have order 2, so must also be 2-cycles themselves. So we have:

$$\begin{aligned}\phi((1\ 2)) &= (a_1\ a_2) \\ \phi((2\ 3)) &= (b_2\ b_3)\end{aligned}$$

for some two element sets $\{a_1, a_2\}$ and $\{b_2, b_3\}$. Since they don't commute, the two element sets must have exactly one element in their intersection $\{a_1, a_2\} \cap \{b_2, b_3\}$, which we call $\{a\}$. After rearranging, set $a = a_2$ and set the unique elements in $\{a_1, a_2\} - \{a_2\}$ and $\{b_2, b_3\} - \{a_2\}$ a_1 and a_3 , respectively. We now have $\phi((1\ 2)) = (a_1\ a_2)$ and $\phi((2\ 3)) = (a_2\ a_3)$. Moreover, $a_1 \neq a_2 \neq a_3$, and clearly, since $\{a_1, a_2\} \neq \{a_2, a_3\}$, we have $a_1 \neq a_3$ as well. So we have shown that ϕ acts as an inner automorphism on $(1\ 2)$ and $(2\ 3)$, and we're done. $\square$

More generally, we have the following result for automorphisms of S_n .

Lemma 9. Let ϕ be an automorphism of S_n . If ϕ sends 2-cycles to 2-cycles, then ϕ must be an inner automorphism.

Proof Outline: Since $(1\ 2), (2\ 3), \dots, (n-1\ n)$ generates S_n (why?), we'll show that ϕ acts as a inner automorphism on these 2-cycles. First, as before, we get $\phi((1\ 2)) = (a_1\ a_2)$ and $\phi((2\ 3)) = (b_2\ b_3)$. And by similar arguments, we get that $\phi((1\ 2)) = (a_1\ a_2)$ and $\phi((2\ 3)) = (a_2\ a_3)$ for some injective function $a : [3] \hookrightarrow [n], i \mapsto a_i$.

Assume $3 \leq k < n$, and we have constructed $a : [k] \hookrightarrow [n], i \mapsto a_i$ such that

$$\begin{aligned}\phi((1\ 2)) &= (a_1\ a_2) \\ \phi((2\ 3)) &= (a_2\ a_3) \\ &\vdots \\ \phi((k-1\ k)) &= (a_{k-1}\ a_k)\end{aligned}$$

We want to extend this to $a : [k+1] \hookrightarrow [n]$ such that $\phi((k\ k+1)) = (a_k\ a_{k+1})$. Note $\phi((k\ k+1)) = (b_k\ b_{k+1})$ since ϕ sends 2-cycles to 2-cycles. Now, $(k\ k+1)$ commutes with $(1\ 2), (2\ 3), \dots, (k-2\ k-1)$, so $b_k, b_{k+1} \notin \{a_1, a_2, \dots, a_{k-1}\}$. The two cycles $(k-1\ k)$ and $(k\ k+1)$ do not commute, so $\{b_k, b_{k+1}\} \cap \{a_{k-1}, a_k\}$ must be a singleton. Since none of b_k, b_{k+1} can be a_{k-1} , it must be a_k . We can call (after rearranging) b_k, b_{k+1} a_k and a_{k+1} . Since $a_{k+1} \neq a_1, \dots, a_{k-1}$ and $a_{k+1} \neq a_k$ (why?), we're done. $\square$

Homework 4 Problem 9: Find all Sylow subgroups of S_n for $n \leq 5$.

Hints: The hard part is finding the Sylow 2-subgroups of S_4 (and S_5 , but it follows easily from the S_4 case). Note all Sylow 2-subgroups are conjugate to each other, so it's enough to find 1 Sylow subgroup.

We know $|S_4| = 4! = 2^3 \cdot 3$. Consider $\sigma := (1\ 2\ 3\ 4)$, and $\langle \sigma \rangle = \{\text{id}, (1\ 2\ 3\ 4), (1\ 3)(2\ 4), (1\ 4\ 3\ 2)\}$. It's not a Sylow 2-subgroup yet, but it can be extended to one. We'll add $\tau := (1\ 3)$ to this group (inspired by one of the elements being $(1\ 3)(2\ 4)$). Now,

$$\begin{aligned} \langle \sigma, \tau \rangle &\supseteq \{\text{id}, \sigma, \sigma^2, \sigma^3, \tau, \tau\sigma, \tau\sigma^2, \tau\sigma^3\} \\ &= \{\text{id}, (1\ 2\ 3\ 4), (1\ 3)(2\ 4), (1\ 4\ 3\ 2), \\ &\quad (1\ 3), (1\ 3)(1\ 2\ 3\ 4) = (1\ 2)(3\ 4), (1\ 3)(1\ 3)(2\ 4) = (2\ 4), (1\ 3)(1\ 4\ 3\ 2) = (1\ 4)(2\ 3)\} \\ &:= X \end{aligned}$$

Clearly, X (as defined as above) has 8 elements. It turns out, X is exactly $\langle \sigma, \tau \rangle$, and hence a Sylow 2-subgroup of S_4 . To see this, note σ and τ satisfy the relation $\sigma^4 = \tau^2 = \text{id}$ and $\tau\sigma\tau^{-1} = \sigma^{-1}$ (check this yourself!). Since $\langle \sigma, \tau \rangle$ consists of all $s_1 s_2 \cdots s_k$ where $s_i \in \{\sigma^{\pm 1}, \tau\sigma^{\pm 1}\}$, we can apply induction on k to show that each $s_1 s_2 \cdots s_k$ must be of the form $\tau^0 \text{ or } 1 \sigma^j$ where $j = 0, 1, 2, 3$. We do this by "sliding" τ to the front using the relation $\tau\sigma\tau^{-1} = \sigma^{-1}$ (details are left as an exercise!). $\square$

Remark. We can go further and show that the subgroup described as above is in fact, the dihedral group D_8 . In fact, if we have a square with vertices labeled 1, 2, 3, 4 in the counterclockwise order, then we can even see which permutation corresponds to which symmetry of the square. Details are left as an exercise!

For the remainder of this section (not covered during discussion), we will properly define the sign of a permutation. In the notes, the definition of a sign is given as follows:

Definition 10 (Sign of a Permutation, as in the Notes). For a $\sigma \in S_n$, $\text{sgn}(\sigma) = \det(\delta_{\sigma(i)j})$.

The issue with the definition is that the definition of $\det$ already uses the sgn . Hence, this definition is circular (even if we use other definition of a determinant, such as the product of eigenvalues counted to algebraic multiplicity, it's not particularly illuminating). Instead, we'll define it using its action on certain polynomial. First, a well-known ring-theoretic fact:

Theorem 10. Consider the ring $\mathbb{Z}[X_1, X_2, \dots, X_n]$ of polynomial (in integer coefficients) of n free variables. We have $S_n \curvearrowright \{X_1, X_2, \dots, X_n\}$ by $\sigma \cdot X_i = X_{\sigma(i)}$. This induces a unique homomorphism for each $\sigma \in S_n$, $\mathbb{Z}[X_1, X_2, \dots, X_n] \rightarrow \mathbb{Z}[X_1, X_2, \dots, X_n], \sigma \cdot P(X_1, X_2, \dots, X_n) = P(X_{\sigma(1)}, X_{\sigma(2)}, \dots, X_{\sigma(n)})$.

Remark. Even though, this seems intuitive, the details of making sure that $P(X_1, X_2, \dots, X_n) \mapsto P(X_{\sigma(1)}, X_{\sigma(2)}, \dots, X_{\sigma(n)})$ is a well-defined ring homomorphism is a bit subtle, and are covered in 210B.

We're now ready to define $\text{sgn}(\sigma)$:

Definition 11 (Sign of a Permutation). Let $\Delta = \prod_{i < j} (X_i - X_j) \in \mathbb{Z}[X_1, X_2, \dots, X_n]$. Then, clearly $\sigma \cdot \Delta = \pm 1 \Delta$ (why?). We define $\text{sgn}(\sigma)$ as the unique $\epsilon \in \{\pm 1\}$ such that $\sigma \cdot \Delta = \epsilon \Delta$.

Proposition 1. For any $\sigma, \tau \in S_n$, $\text{sgn}(\tau \circ \sigma) = \text{sgn}(\tau)\text{sgn}(\sigma)$.

Proof. We just compute $(\tau \circ \sigma) \cdot \Delta$

$$\begin{aligned} (\tau \circ \sigma) \cdot \Delta &= \tau \cdot (\sigma \cdot \Delta) \\ &= \tau \cdot (\text{sgn}(\sigma)\Delta) \\ &= \text{sgn}(\sigma)\tau \cdot \Delta \\ &= \text{sgn}(\sigma)\text{sgn}(\tau)\Delta \end{aligned}$$

□

Corollary 4. For $\sigma, \tau \in S_n$, we have $\text{sgn}(1) = 1$, $\text{sgn}(\sigma)\text{sgn}(\sigma^{-1}) = 1$ (in particular, $\text{sgn}(\sigma) = \text{sgn}(\sigma^{-1})$), and $\text{sgn}(\sigma \circ \tau \sigma^{-1}) = \text{sgn}(\tau)$.

Proof. $\sigma(1) = 1$ is trivial. Note $1 = \text{sgn}(\sigma \circ \sigma^{-1}) = \text{sgn}(\sigma)\text{sgn}(\sigma^{-1})$. Now, for the conjugation invariance, we compute $\text{sgn}(\sigma \circ \tau \sigma^{-1}) = \text{sgn}(\sigma)\text{sgn}(\tau)\text{sgn}(\sigma^{-1}) = \text{sgn}(\tau)$. □

We still have not shown that sgn is nontrivial. We'll show that the sign of a 2-cycle is -1 .

Proposition 2. For any 2-cycle $(a \ b)$, $\text{sgn}((a \ b)) = -1$.

Proof. By conjugation, it's enough to show $\text{sgn}((1 \ 2)) = -1$. We now compute $(1 \ 2) \cdot \Delta$ as follows. First, Δ can be broken down into the following

$$\begin{aligned} \Delta &= \prod_{i < j} (X_i - X_j) \\ &= (X_1 - X_2) \prod_{j > 2} (X_1 - X_j) \prod_{j > 2} (X_2 - X_j) \prod_{2 < i < j} (X_i - X_j) \end{aligned}$$

The 2-cycle $(1 \ 2)$ sends $(X_1 - X_2)$ to $(X_2 - X_1) = -(X_1 - X_2)$, sends $\prod_{j > 2} (X_1 - X_j)$ and $\prod_{j > 2} (X_2 - X_j)$ to each other, and fixes $\prod_{2 < i < j} (X_i - X_j)$. Hence, we see that $(1 \ 2) \cdot \Delta = -\Delta$, so $\text{sgn}((1 \ 2)) = -1$. □

Remark. It's now clear that $\text{sgn}(\sigma)$ is $(-1)^k$ where k is the number of 2-cycles in its 2-cycle decomposition. Obviously, since sgn itself is well-defined, it follows that the number k of 2-cycles must be the independent modulo 2 of a specific 2-cycle decomposition.

Discussion 6 - Semidirect Products, Short Exact Sequences of Groups

Here's a qual problem which I planned on going over in week 4, but did not have the time to.

Fall 2021 Problem 3: Show that there are no simple groups of order $24p$, where p is a prime number greater than 11.

Proof Outline: In the $p \neq 23$ case, $n_p \mid 24$ and $n_p \equiv 1 \pmod{p}$. This forces $n_p = 1$, so the p -Sylow subgroup is normal.

Assume $p = 23$. In this case, $n_p = 24$ or 1 . Assume $n_p = 24$ since we'd be done in the $n_p = 1$ case. We can now, look at n_3 . We know $n_3 \mid 2^3 \cdot 23$, and $n_3 \equiv 1 \pmod{3}$.

Case (i): $n_3 = 1$, we're done.

Case (ii): $n_3 = 2^2$. In this case, $n_3 = [G : N_G(P_3)] = 2^2$, so by Cayley normal subgroup theorem, there exists $N \leq N_G(P_3)$, $N \trianglelefteq G$ such that $[G : N] \mid 8!$. Convince yourself that $N \neq \{e\}$.

Case (iii): $n_3 = 2^2 \cdot 23 \geq 46$. In this case, there are way too many 3 and 23 Sylow subgroups to fit into G . $\square$

Here's a homework problem on groups of order p^3 .

HW 5 Problem 4: Let G be a non-abelian group of order p^3 . Show that $Z(G) = [G, G]$.

Hints: First, show that $|Z(G)| = p$. Then, use this to show that $[G, G] \leq Z(G)$ (recall there is an equivalent condition for $[G, G]$ to be contained in a normal subgroup N). So from here, we see that $[G, G]$ is either trivial or all of $Z(G)$. But since G is non-abelian, $[G, G]$ cannot be trivial. $\square$

We now recall the definition of a semi-direct product.

Definition 12 (Internal Semi-direct Product). Let N, H be subgroups of a group G and $N \trianglelefteq G$. If $N \cap H = \{e\}$, then NH is also a subgroup of G , called the (*internal*) *semi-direct product* of N and H . If $G = NH$, then we say G is a semi-direct product of N and H .

Remark. The group operation on NH is induced from G , so it looks as follows. For any $n_1, n_2 \in N$ and $h_1, h_2 \in H$, $n_1 h_1 n_2 h_2 = n_1 (h_1 n_2 h_1^{-1}) h_1 h_2$. Inspired by this, we will provide a more abstract definition of a semi-direct product below.

Definition 13 (External Semi-direct Product). Let N and H be two groups, and suppose we have a homomorphism $\phi : H \rightarrow \text{Aut}(N)$. We define the *external semi-direct product* of N and H , denoted $N \rtimes_{\phi} H$ (or just $N \rtimes H$ if we suppress ϕ) to be the following group. As a set, $N \rtimes_{\phi} H = \{(n, h) \mid n \in N, h \in H\}$. The group operation is given by $(n_1, h_1) \cdot (n_2, h_2) = (n_1 \phi(h_1)(n_2), h_1 h_2)$.

As the names suggest, the two semi-direct products are equivalent in a precise way we'll formulate below.

Theorem 11. There is a correspondence between internal and external semi-direct products. On one hand, given an internal semidirect product between $N \trianglelefteq G$ and $H \leq G$, NH , we have an induced homomorphism $\phi : H \rightarrow \text{Aut}(N)$ given by $\phi(h) : n \rightarrow hnh^{-1}$. Moreover, $NH \cong N \rtimes_{\phi} H$. That is, the internal semi-direct product is isomorphic to the external semi-direct product with the induced homomorphism.

On the other hand, given an external semi-direct product $G := N \rtimes_{\phi} H$, we have an embedding $N \cong N \times \{e\} \trianglelefteq N \rtimes H$, $H \cong \{e\} \times H \leq N \rtimes H$, and $G = N \rtimes H = (N \times \{e\})(\{e\} \times H)$. Moreover, $(e, h) \cdot (n, e) \cdot (e, h)^{-1}$ is given by $(\phi(h)(n), e)$. In other words, external semi-direct products constructs a larger group G containing both N and H in which G is the internal semi-direct product of N and H .

Proof Outline: Suppose we're given $G = NH$. Then, note first, since $N \cap H = \{e\}$, $N \times H \rightarrow NH, (n, h) \mapsto nh$ is a bijection (why?). The multiplication on NH is given by $(n_1 h_1)(n_2 h_2) = n_1(h_1 n_2 h_1^{-1})h_1 h_2$ so the multiplication on $N \rtimes_{\phi} H$ mimics the multiplication on NH (that is, the map $(n, h) \mapsto nh$ is a homomorphism).

Now, suppose we have $N \rtimes H$. Clearly, $N \rightarrow N \times \{e\} \leq N \rtimes H$ and $H \rightarrow \{e\} \times H \leq N \rtimes H$ are injections onto the images, whose intersection is clearly $\{e\} \times \{e\}$. Let's check they are also homomorphisms: $(n_1, e) \cdot (n_2, e) = (n_1 \phi(e)(n_2), e \cdot e) = (n_1 n_2, e)$, and $(e, h_1) \cdot (e, h_2) = (e \phi(h_1)(e), h_1 h_2) = (e, h_1 h_2)$.

Now, also, $(n, e) \cdot (e, h) = (n \phi(e)(e), eh) = (n, h)$, so in fact, $N \rtimes H = (N \times \{e\})(\{e\} \times H)$. Finally, to check the last claim, $(e, h) \cdot (n, e) \cdot (e, h)^{-1} = (\phi(h)(n), h) \cdot (e, h^{-1}) = (\phi(h)(n) \phi(h)(e), hh^{-1}) = (\phi(h)(n), e)$. So we see that conjugation by (e, h) on $N \times \{e\}$ mimics $\phi(h)$ on N . $\square$

Homework 5 Problem 9: Show that $\mathbb{Z}/n\mathbb{Z} \rtimes (\mathbb{Z}/n\mathbb{Z})^{\times} \hookrightarrow S_n$ where the semi-direct product is given by $(\mathbb{Z}/n\mathbb{Z})^{\times} \cong \text{Aut}(\mathbb{Z}/n\mathbb{Z}), k \mapsto k \cdot (-)$.

Hints: We want $G := \mathbb{Z}/n\mathbb{Z} \rtimes (\mathbb{Z}/n\mathbb{Z})^{\times} \hookrightarrow S_n$, which corresponds to a group action $G \curvearrowright [n] \cong \mathbb{Z}/n\mathbb{Z}$ (we see that $\mathbb{Z}/n\mathbb{Z}$ plays nicer with G than $[n]$).

Since $G = (\mathbb{Z}/n\mathbb{Z} \times \{1\})(\{0\} \times (\mathbb{Z}/n\mathbb{Z})^{\times})$, it's enough to see how each groups $\mathbb{Z}/n\mathbb{Z}$ and $(\mathbb{Z}/n\mathbb{Z})^{\times}$ acts on $\mathbb{Z}/n\mathbb{Z}$. There are obvious actions $k \cdot m = k + m$ and $l \cdot m = lm$. Combine them to induce an action all of $G = \mathbb{Z}/n\mathbb{Z} \rtimes (\mathbb{Z}/n\mathbb{Z})^{\times}$. $\square$

Fall 2020 Problem 2: Show that groups of order $231 = (3)(7)(11)$ are semi-direct products and show that there are exactly two such groups up to isomorphism.

Proof. Let's analyse the Sylow subgroups. Let n_p be the number of Sylow p -subgroups. From the third Sylow's theorem, we get $n_7 \mid 33, n_7 \equiv 1 \pmod{7}$ and $n_{11} \mid 21, n_{11} \equiv 1 \pmod{11}$. Hence, $n_7 = n_{11} = 1$, so both Sylow 7 and 11 subgroups are normal. Let P_7 and P_{11} be the Sylow 7 and 11 subgroups, respectively. Then, since $P_7, P_{11} \trianglelefteq G$ and $P_7 \cap P_{11} = \{e\}$, $P_7 P_{11} \trianglelefteq G$ (why?). Moreover, since P_7 and P_{11} are both normal in $H := P_7 P_{11}$, $H \cong P_7 \times P_{11} \cong \mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/11\mathbb{Z} \cong \mathbb{Z}/77\mathbb{Z}$.

Now, take P_3 a Sylow 3-subgroup of G . Then, clearly, $H \trianglelefteq G$ and $P_3 \cap H = \{e\}$. So $G = H P_3 \cong \mathbb{Z}/77\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$, with the induced homomorphism $\phi : \mathbb{Z}/3\mathbb{Z} \rightarrow \text{Aut}(\mathbb{Z}/77\mathbb{Z})$. This shows G is a semi-direct product, and moreover, the two isomorphism classes correspond to the case where ϕ is trivial and where ϕ is nontrivial. We leave it up to you as an exercise to show that all non-trivial ϕ induce the same isomorphism class of $\mathbb{Z}/77\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$ (use HW 5 Problems 7 and 8). $\square$

Finally, we'll review a category theoretic characterization of semi-direct products, and prove an analogous characterization for direct products. That will also segue nicely into one of the qual problems.

Recall first, the definition of an exact sequence:

Definition 14 (Exact Sequence of Groups). Let

$$\cdots \rightarrow G_{i-1} \xrightarrow{f_{i-1}} G_i \xrightarrow{f_i} G_{i+1} \xrightarrow{f_{i+1}} G_{i+2} \rightarrow \cdots$$

be a sequence of groups with homomorphisms. We say the sequence is *exact* if $\text{Im}(f_{i-1}) = \ker(f_i)$ for all i . In particular, we call the exact sequence of the form below

$$1 \rightarrow N \rightarrow G \rightarrow H \rightarrow 1$$

a short exact sequence.

Short exact sequences correspond to $N \trianglelefteq G$ and the quotient G/N .

Proposition 3. Given an exact sequence $1 \rightarrow N \xrightarrow{j} G \xrightarrow{\pi} H \rightarrow 1$, j embeds N into the normal subgroup (which we'll also call N , and hence $N \trianglelefteq G$) $\ker \pi$, and $\pi : G \twoheadrightarrow H$ induces an isomorphism $G/N \rightarrow H$. Conversely, given a normal subgroup $N \trianglelefteq G$, the sequence $1 \rightarrow N \rightarrow G \rightarrow G/N \rightarrow 1$ is exact.

Given a normal subgroup $N \trianglelefteq G$ and the quotient, G/N , we don't necessarily have a way of reconstructing G as a semi-direct product. The ones where we can correspond to a very special kind of exact sequence.

Theorem 12. The following are equivalent:

- (I) The exact sequence $1 \rightarrow N \xrightarrow{j} G \xrightarrow{\pi} H \rightarrow 1$ *splits on the right*. That is, there exists a section $s : H \rightarrow G$ such that $\pi \circ s = \text{id}_H$.
- (II) There is an isomorphism $\phi : G \xrightarrow{\cong} N \rtimes H$ which makes the following diagram commute:

$$\begin{array}{ccccccc} 1 & \rightarrow & N & \xrightarrow{j} & G & \xrightarrow{\pi} & H \rightarrow 1 \\ & & \downarrow \text{id} & & \downarrow \exists \phi \downarrow \cong & & \downarrow \text{id} \\ 1 & \rightarrow & N & \rightarrow & N \rtimes H & \rightarrow & H \rightarrow 1 \end{array}$$

So, semi-direct products correspond to right split exact sequences. It turns out, direct products correspond to *left* split exact sequences.

Theorem 13. The following are equivalent:

- (I) The short exact sequence $1 \rightarrow N \xrightarrow{j} G \xrightarrow{\pi} H \rightarrow 1$ *splits on the left*. That is, there exists $s : G \rightarrow N$ such that $s \circ j = \text{id}_N$.
- (II) There is an isomorphism $\phi : G \xrightarrow{\cong} N \times H$ which makes the following diagram commute:

$$\begin{array}{ccccccc} 1 & \rightarrow & N & \xrightarrow{j} & G & \xrightarrow{\pi} & H \rightarrow 1 \\ & & \downarrow \text{id} & & \downarrow \exists \phi \downarrow \cong & & \downarrow \text{id} \\ 1 & \rightarrow & N & \rightarrow & N \times H & \rightarrow & H \rightarrow 1 \end{array}$$

Proof. The (II) $\implies$ (I) part is trivial and is left as an exercise. Assume (I), and pick a left splitting map $s : G \rightarrow N$ such that $s \circ j = \text{id}_N$. Define $\phi : G \rightarrow N \times H$ by $\phi(g) = (s(g), \pi(g))$. We'll show that ϕ is an isomorphism. Suppose $g \in \ker \phi$. Then, $\pi(g) = e$, so $g \in j(N)$, i.e. $g = j(n)$ for a unique value of $n \in N$. Then, $s(g) = s \circ j(n) = n = e$, so $n = e$. Hence, g must be the identity as well.

Now, take any $(n, h) \in N \times H$. Choose any $g \in G$ such that $\pi(g) = h$. Now, for any $k \in N$, $\pi(gj(k)) = \pi(g) = h$. We want to choose $k \in N$ such that $s(gj(k)) = n$. But this is equivalent to $s(g)k = n$, i.e. $k = s(g)^{-1}n$. So the element $gs(g)^{-1}n$ gets sent to (n, h) under ϕ .

Finally, checking that ϕ makes the diagram above commute is left as an exercise. $\square$

Definition 15 (Outer Automorphism Group). Recall that given a group H , $\text{Inn}(H) \trianglelefteq \text{Aut}(H)$. The *outer automorphism group* is defined as the quotient $\text{Out}(H) := \text{Aut}(H)/\text{Inn}(H)$.

Theorem 14 (Schreier Conjecture). If H is a finite simple group, then $\text{Out}(H)$ is solvable.

Remark. Even though, Schreier Conjecture is proven and is listed as a theorem here, I still prefer to call it a "conjecture" because the proof relies on the classification of finite simple groups. That is, there is no known "elementary" proof of this fact.

This segues nicely into one of the qual problems.

Fall 2021 Problem 9: Use Schreier conjecture to show that if G is a finite group such that both N and G/N are simple non-abelian, then $G \cong N \times G/N$.

Proofs will have to wait until later discussions (or if we don't get to it, I'll write it here).

Discussion 7 - Free Groups and Presentation of Groups

For today's discussion, we will review the properties of free groups. In week 0, I gave an informal description of groups as generated by automorphisms of certain structures, together with composition. This time, we're constructing groups from "scratch" or "artificially". Informally speaking, free groups are groups generated by letters without any relations imposed on the letters. The construction of free groups captures this idea:

Definition 16 (Construction of Free Groups). Let X be a set (of letters), and define the set $X \cup X^{-1}$ of its letters and their formal inverses. A *word* in X (and its inverse set X^{-1}) is a (possibly empty) sequence of letters $x_1x_2 \cdots x_n$ where $x_i \in X \cup X^{-1}$. Denote $W(X)$ the set of all words in X . We can define a binary operation $(x_1x_2 \cdots x_n) \cdot (y_1y_2 \cdots y_m) = x_1x_2 \cdots x_ny_1y_2 \cdots y_m$ by concatenation.

We set $\sim$ to be the equivalence relation generated by the relations $uaa^{-1}v \sim uv$ for $a \in X \cup X^{-1}$. I.e. this captures the idea that a letter and its inverse should cancel each other out, and in fact, the only things that cancel each other out.

The free group $F(X)$ is defined to be the set of equivalence classes $F(X) := W(X)/\sim = \{[w]_\sim \mid w \text{ is a word in } X\}$ together with the binary operation induced from concatenation $[w_1]_\sim \cdot [w_2]_\sim = [w_1 \cdot w_2]_\sim$ (the operation is well-defined since reducing a word does not change the equivalence class of concatenation).

There is another way to state that the free group on letters X is a group of words without any relation on different letters. We show that the free group $F(X)$ can be sent to any group G as long as we specify where we send each letter $x \in X$ to (whatever relations the images in G might have).

Theorem 15. Let $F(X)$ be the free group on letters X . Then, $F(X)$ satisfies the following universal property: for every group G and $\forall \phi : X \rightarrow G$ a function, there exists unique homomorphism $\phi : F(X) \rightarrow G$ such that $\phi([x]_\sim) = \phi(x)$. In other words, if we define $j : X \rightarrow F(X)$ by $x \mapsto [x]_\sim$, then the following diagram commutes:

$$\begin{array}{ccc} X & & \\ \downarrow j & \searrow \forall \phi & \\ F(X) & \xrightarrow{\exists! \phi} & G \end{array}$$

Remark. The induced homomorphism sends $[x_1x_2 \cdots x_n]$ to $\phi(x_1)\phi(x_2) \cdots \phi(x_n)$. Here, if $a = x^{-1} \in X^{-1}$, we mean by $\phi(a)$, $\phi(x)^{-1}$.

Proposition 4. The universal property characterizes the free group uniquely. More precisely, given another $G(X)$ and $\tilde{j} : X \rightarrow G(X)$ satisfying the universal property of free groups, there exists unique isomorphism $f : F(X) \xrightarrow{\cong} G(X)$ such that $f \circ j = \tilde{j}$. In other words, the following diagram commutes

$$\begin{array}{ccc} F(X) & \xrightarrow[\exists! f]{\cong} & G(X) \\ & \nwarrow j \quad \nearrow \tilde{j} & \\ & X & \end{array}$$

Proof. By applying universal properties on both groups $F(X)$ and $G(X)$, we have unique induced maps from $\tilde{J} : F(X) \rightarrow G(X)$ and $J : G(X) \rightarrow F(X)$ induced from $\tilde{j} : X \rightarrow G(X)$ and $j : X \rightarrow F(X)$, respectively. In other words, the following diagram commutes:

$$\begin{array}{ccc} F(X) & \xleftarrow[\exists! J]{\exists! \tilde{J}} & G(X) \\ & \nwarrow j \quad \nearrow \tilde{j} & \\ & X & \end{array}$$

We will show that $J \circ \tilde{J} : F(X) \rightarrow F(X)$ is the identity.

$$X \xrightarrow{j} F(X) \xleftarrow{J \circ \tilde{J}, \text{id}}$$

Note $J \circ \tilde{J} \circ j = J \circ \tilde{j} = j = \text{id} \circ j$. So both $J \circ \tilde{J}$ and id commute with j , but by uniqueness, $J \circ \tilde{J} = \text{id}$. Similarly, one can show $\tilde{J} \circ J = \text{id}$ as well. $\square$

Exercise 7. Show without using the construction (and only using the universal property), that the free group $F(X)$ is generated by $j(X)$ (hints: use the universal property to map $F(X)$ into the subgroup generated by $j(X)$. It will turn out to be the inverse of the inclusion map).

Remark. Since the universal property characterizes the free group up to unique isomorphism, some authors *define* the free group as the group $F(X)$ (together with a map $j : X \rightarrow F(X)$) satisfying the universal property. From there, the construction is no longer the definition, but a proof of existence (more accurately, construction together with the proof of universal property is). We just proved the uniqueness up to unique isomorphism.

In lecture, we showed that given a word w , there is a unique reduced word $\tilde{w}$ with $[\tilde{w}] = [w]$. The proof is combinatorial and uses induction on lengths. We will show you a proof which uses the universal property and is a bit more algebraic.

Proposition 5. Given $[w] \in F(X)$, there exists a unique reduced word $\tilde{w}$ with $[\tilde{w}] = [w]$.

Proof. Set $R(X)$ the set of all reduced words in X . Then, define $\phi : X \rightarrow S_{R(X)}$ by $\phi(x) = L_x$ where $L_x : \begin{cases} x^{-1}w \mapsto w \\ u \mapsto xu \end{cases}$ if u does not start with x^{-1} .

We can check that since $L_{x^{-1}}$ (when defined similarly) is the inverse, so L_x really does lie in $S_{R(X)}$. By the universal property, this induces a unique homomorphism $\phi : F(X) \rightarrow S_{R(X)}$, $[x_1 x_2 \cdots x_n]_{\sim} \mapsto L_{x_1} \circ L_{x_2} \circ \cdots \circ L_{x_n}$. This also induces a group action $F(X) \curvearrowright R(X)$ by $[w]_{\sim} \cdot u = \phi([w]_{\sim})(u)$.

Claim. If w is a reduced word, then $[w]_{\sim} \cdot \emptyset = w$.

This is the place where we apply induction on the length of the reduced word w (which is an easy exercise left to you). Applying the claim, if w_1 and w_2 are the reduced words in $[w]$, then $[w] \cdot \emptyset = [w_1] \cdot \emptyset = [w_2] \cdot \emptyset$, so we get $w_1 = w_2$. $\square$

In previous lectures and discussions, we gave informal descriptions of a group generated by certain elements with imposed relations. We will make this idea more precise:

Definition 17 (Presentation of a Group). Let X be a set and $R \subset W(X)$. Then,

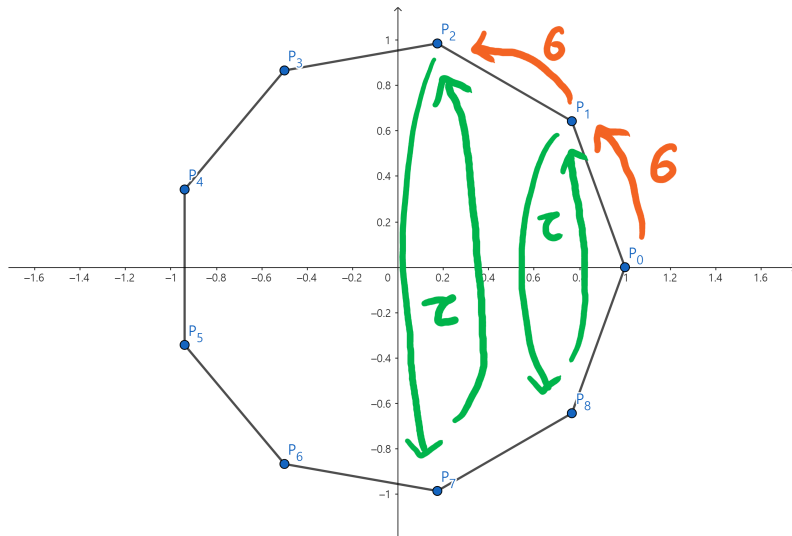
$$\langle X \mid R \rangle := F(X) / \langle\langle R \rangle\rangle$$

where $\langle\langle R \rangle\rangle$ is the normal subgroup of $F(X)$ generated by R . That is, the group $\langle X \mid R \rangle$ is the free group $F(X)$ quotiented by the relation R we wish to impose on $F(X)$.

Below, we provide (and prove) a presentation of the dihedral group D_{2n} . Recall that the dihedral group D_{2n} is the group of isometries of the plane keeping an n -Gon invariant.

Proposition 6. For $n \geq 1$, $D_{2n} \cong \langle \sigma, \tau \mid \sigma^n = \tau^2 = e, \tau\sigma\tau = \sigma^{-1} \rangle$.

Proof. Draw the n -Gon in the plane as shown in the picture, with vertices labelled $0, 1, \dots, n-1 \in \mathbb{Z}/n\mathbb{Z}$ positioned counter-clockwise. Let $\sigma \in D_{2n}$ be the counter-clockwise rotation sending 0 to 1, 1 to 2, ..., $n-1$ to 0, and let $\tau \in D_{2n}$ be the reflection across the line of symmetry passing through 0 (this flips 1 to $n-1$, 2 to $n-2$, etc.)



A drawing of an $n = 9$ -Gon on a plane. The rotation σ rotates the 9-Gon counter-clockwise one vertices. The reflection τ reflects across the axis of symmetry along 0, flipping 1 and $n-1$, 2 and $n-2$, etc.

More algebraically, $\sigma : m \mapsto m + 1$ and $\tau : m \mapsto -m$.

Claim. The elements σ, τ generates D_{2n} . In fact, every element $f \in D_{2n}$ is of the form either σ^k or $\sigma^k \tau$ (for some $k = 0, 1, \dots, n-1$).

Proof of the Claim. Let $f \in D_{2n}$, and $k = f(0)$. Then, $f(1)$ is either the vertex $f(0) + 1 = k + 1$ or $f(0) - 1 = k - 1$.

Case 1: $f(1) = k + 1$.

In that case, $f(2) = k + 2, f(3) = k + 3, \dots, f(m) = m + k, \dots$ (formal proof uses induction on m).

Case 2: $f(1) = k - 1$.

In that case, $f(2) = k - 2, f(3) = k - 3, \dots, f(m) = k - m, \dots$ (again, use induction on m to formalize this).

In case 1, $f(m) = m + k = \sigma^k(m)$. In case 2, $f(m) = k - m = \sigma^k(-m) = \sigma^k \circ \tau(m)$. This concludes the proof since $f = \sigma^k$ or $f = \sigma^k \tau$. $\square$

Now, consider the free group $F(\sigma, \tau)$ on the letters σ and τ , and use the universal property to construct the unique homomorphism

$$\begin{aligned} \phi : F(\sigma, \tau) &\twoheadrightarrow D_{2n} \\ \sigma &\mapsto \sigma(m \mapsto m + 1) \\ \tau &\mapsto \tau(m \mapsto -m) \end{aligned}$$

By the claim, ϕ is surjective. Moreover, we leave it as an exercise to check that $\sigma^n, \tau^2, \tau\sigma\tau\sigma \in \ker \phi$. By the first isomorphism theorem, this induces a unique homomorphism

$$\bar{\phi} : G := \langle \sigma, \tau \mid \sigma^n = \tau^2 = e, \tau\sigma\tau = \sigma^{-1} \rangle = F(\sigma, \tau) / \langle \langle \sigma^n, \tau^2, \tau\sigma\tau\sigma \rangle \rangle \twoheadrightarrow D_{2n}$$

Convince yourself that every element of G can also be written as σ^k or $\sigma^k \tau$ (this can be done by induction on the length of the word $x_1 x_2 \cdots x_m$ with $x_i \in \{\sigma^{\pm 1}, \tau^{\pm 1}\}$).

Since $\bar{\phi}$ sends each σ^k and $\sigma^k \tau$ to $\sigma^k, \sigma^k \tau$, which are distinct in D_{2n} , the original $\sigma^k, \sigma^k \tau$ in G must also be distinct as well. Moreover, $\bar{\phi}$ sends those distinct $\sigma^k, \sigma^k \tau$ in G to distinct $\sigma^k, \sigma^k \tau$ in D_{2n} , so $\bar{\phi}$ is injective. Hence, $\bar{\phi}$ is an isomorphism. $\square$

HW 6 Problem 7: Let $G = \langle a, b \mid w^3 = e \rangle$ where the w ranges over all possible words w in $F(a, b)$. Show that $|G| = 27$.

Hints: As given in the hints, first set $N = \langle ab, ba \rangle$, and show that ab and ba commute. This implies N is abelian. Note since $\forall w \in N, w^3 = e$, we can define a scalar multiplication $\mathbb{Z}/3\mathbb{Z} \times N \rightarrow N$ by $k \cdot w = w^k$ (the fact that $w^3 = e$ implies that w^k is independent of the representative k and is therefore, well-defined). Hence, N is a 2-generated $\mathbb{Z}/3\mathbb{Z}$ -vector space, so it is either isomorphic to $(\mathbb{Z}/3\mathbb{Z})^2, \mathbb{Z}/3\mathbb{Z}$, or $\{e\}$. We believe N is $(\mathbb{Z}/3\mathbb{Z})^2$, as we will show later.

Now, we also show $N \trianglelefteq G$ before looking at the quotient group G/N . Note $N \trianglelefteq G$ if and only if $N_G(N) = G$ if and only if $a, b \in N_G(N)$. The latter is equivalent to for all $g \in \{ab, ba\}$ (the generating set), $aga^{-1} \in N$ and $bgb^{-1} \in N$. We leave this part as an exercise.

Note in G/N , $abN = N$, so $bN = a^{-1}N$. So G/N is cyclically generated by aN , whose order is either 3 or 1. We believe $\text{ord}(aN) = 3$ (and hence $G/N \cong \mathbb{Z}/3\mathbb{Z}$), as we will show below.

Consider the homomorphism $F(a, b) \xrightarrow{\phi} \mathbb{Z}/3\mathbb{Z}$ induced by $a \mapsto 1, b \mapsto 2$. Show that this descends all the way to $G/N \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ by first, descending this to $\bar{\phi} : G := F(a, b)/\langle\langle w^3, w \in F(a, b) \rangle\rangle \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$, then, further descending this to $\overline{\bar{\phi}} : G/N \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$. The first descent can be done by showing $\phi(w^3) = 3 \cdot \phi(w) = 0$, and hence, $w^3 \in \ker \phi$. The second descent can be done by showing $ab, ba \in \ker \phi$. The details are left to you.

Once we have the induced homomorphism $\overline{\bar{\phi}} : G/N \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}, aN \mapsto \phi(a) = 1 \neq 0$, we see that aN cannot be the identity because the identity must get mapped to the identity. Further note that this is in fact, an isomorphism from G/N to $\mathbb{Z}/3\mathbb{Z}$.

We have a short exact sequence $1 \rightarrow N \hookrightarrow G \xrightarrow{\pi} G/N \rightarrow 1$ with a splitting map $aN \mapsto a$. Hence, $G = N\langle a \rangle \cong N \rtimes \langle a \rangle$ with $N \cap \langle a \rangle = \{e\}$. We wish to show that $N \cong (\mathbb{Z}/3\mathbb{Z})^2$, and therefore, $G \cong (\mathbb{Z}/3\mathbb{Z})^2 \rtimes \mathbb{Z}/3\mathbb{Z}$. We don't know that yet, but if it were true, ab would pair with $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and ba would pair with $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Moreover, a would pair with $\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 1\right)$ in the semi-direct product $(\mathbb{Z}/3\mathbb{Z})^2 \rtimes \mathbb{Z}/3\mathbb{Z}$. And the homomorphism $f : \mathbb{Z}/3\mathbb{Z} \rightarrow \text{Aut}((\mathbb{Z}/3\mathbb{Z})^2) = \text{GL}_2(\mathbb{Z}/3\mathbb{Z})$ defining the semi-direct product would mimic conjugation by a on N (more precisely, $f(k)$ will mimic conjugation by a^k on N).

Observe $a(ba)a^{-1} = ab$ and $a(ab)a^{-1} = a^2ba^2 = (ab)^?(ba)^?$. Hence, $f(1) \in \text{GL}_2(\mathbb{Z}/3\mathbb{Z})$ must be a linear map of the form:

$$f(1) : \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} ? \\ ? \end{pmatrix}$$

Set the matrix $A = \begin{pmatrix} ? & 1 \\ ? & 0 \end{pmatrix}$. Check that $A^3 = I$. This will allow us to show that $f : k \mapsto A^k$ is a well-defined homomorphism, and hence, that $(\mathbb{Z}/3\mathbb{Z})^2 \rtimes \mathbb{Z}/3\mathbb{Z}$ is a valid semi-direct product.

Next, we have to show $G \cong (\mathbb{Z}/3\mathbb{Z})^2 \rtimes \mathbb{Z}/3\mathbb{Z}$, the latter which we just defined. Again, we descend the homomorphism from the free group: define $\phi : F(a, b) \rightarrow (\mathbb{Z}/3\mathbb{Z})^2 \rtimes \mathbb{Z}/3\mathbb{Z}$ by $a \mapsto \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, 1\right)$ and $b \mapsto \left(\begin{pmatrix} ?? \\ ?? \end{pmatrix}, ??\right)$. When done properly, $ab \mapsto \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, 0\right)$ and $ba \mapsto \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}, 0\right)$.

Check that $\forall w \in F(a, b), w^3 \in \ker \phi$. For this, use the fact that for all $(v, k) \in (\mathbb{Z}/3\mathbb{Z})^2 \rtimes \mathbb{Z}/3\mathbb{Z}, (v, k)^3 = ((I + A^{2k} + A^{4k})v, 3k)$. Convince yourself that $I + A + A^2 = 0$ (use $A^3 = I$ and $A - I$ is invertible) to finish $k = 1, 2$ cases.

Once we descend to a homomorphism $\bar{\phi} : G \rightarrow (\mathbb{Z}/3\mathbb{Z})^2 \rtimes \mathbb{Z}/3\mathbb{Z}$, we can restrict to $N \trianglelefteq G$. But we saw $ab \mapsto \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, 0\right)$ and $ba \mapsto \left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}, 0\right)$, which shows us $N \cong (\mathbb{Z}/3\mathbb{Z})^2$ (why?). □

Extra Material: Solvability Qual Problems

This section is on solvability qual problems that we did not get to in discussions.

Let's recall some facts about solvability.

Theorem 16. Let G be a group. Define the derived series $G^{(i)}$ for $i \geq 0$ inductively as follows: $G^{(0)} = G$, and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. Then, the following are equivalent:

- (I) The derived series $G^{(i)}$ stabilizes, that is $G^{(i)} = \{e\}$ for some i .
- (II) There exists a series of subgroups $G = N_0 \supseteq N_1 \supseteq \cdots \supseteq N_n = \{e\}$ such that N_i/N_{i+1} is abelian.

Proof Outline: (I) $\implies$ (II): Note $N_i = G^{(i)}$ is such a series of subgroups.

(II) $\implies$ (I): The condition N_i/N_{i+1} is abelian is equivalent to $[N_i, N_i] \leq N_{i+1}$. With this and induction on $i \geq 0$, we can show that $G^{(i)} \leq N_i$. $\square$

Definition 18 (Solvability). Let G be a group. We say G is solvable if either of the two equivalent conditions (I) and (II) hold true.

Proposition 7. Every simple non-abelian group G is solvable.

Proof. Note $G^{(1)} = [G, G] \trianglelefteq G$. Since G is non-abelian, $[G, G] \neq \{e\}$. So by simplicity, $[G, G] = G$. By induction on i , we can show $G^{(i)} = G$. $\square$

Corollary 5. The alternating group A_n is not solvable if $n \geq 5$.

In fact, the permutation group S_n is not solvable either.

Proposition 8. The permutation group S_n is not solvable for $n \geq 5$.

Proof. Observe $[S_n, S_n] \leq A_n$ since $S_n/A_n \cong \{\pm 1\}$ is an abelian group. Since S_n is non-abelian, $[S_n, S_n] \neq \{e\}$. Since A_n is simple, it forces $[S_n, S_n] = A_n$. Then, by induction on $i \geq 1$, $S^{(i)} = A_n$ for all $i \geq 1$. $\square$

Remark. The fact that S_n is not solvable for $n \geq 5$ is the reason why polynomial equations of degree $n \geq 5$ are not always solvable with radicals. More details in 210B.

We're now ready to tackle some of the qual problems.

Fall 2024 Problem 6: Let G be the group of 3×3 complex matrices of the form

$$\begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

with non-zero entries on the diagonal. (The group operation is matrix multiplication.) Show that G is solvable.

Proof. We leave many of the details as exercises. Take

$$\begin{aligned} N_0 = G, N_1 &= \left\{ \begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ z & y & 1 \end{pmatrix} \mid x, y, z \in \mathbb{C} \right\} \\ N_2 &= \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{pmatrix} \mid c \in \mathbb{C} \right\} \\ N_3 &= \{\text{Id}_3\}. \end{aligned}$$

We can show that

$$G \twoheadrightarrow (\mathbb{C}^\times)^3$$

$$\begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \mapsto (a_{11}, a_{22}, a_{33})$$

is a homomorphism onto the the product of the multiplicative group $\mathbb{C}^\times$, with kernel N_1 . Hence, N_0/N_1 is abelian. Similarly, we can show that

$$N_1 \twoheadrightarrow \mathbb{C}^2$$

$$\begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ z & y & 0 \end{pmatrix} \mapsto (x, y)$$

is a homomorphism from N_1 onto the additive group $\mathbb{C}^2$ with kernel exactly N_2 . Similarly, $N_3 \cong \mathbb{C}$ with the obvious homomorphism, with kernel $\{\text{Id}_3\}$. $\square$

Spring 2019 Problem 1: Let G be a finite solvable group and $1 \neq N \subset G$ be a minimal normal subgroup. Prove that there exists a prime p such that N is either cyclic of order p or a direct product of cyclic groups of order p .

Proof. First, we will show that N is abelian. Observe $[N, N] \trianglelefteq G$ and clearly, $[N, N] \leq N$. Since N is a minimal normal subgroup of G , it follows $[N, N] = 1$ or $[N, N] = N$. The latter cannot happen since then, for all $i \geq 0$, $G^{(i)} \geq N$, contradicting the fact that G is solvable. Hence, $[N, N] = 1$ and equivalently, N is abelian.

Now, for each prime p , consider the map $\phi_p : N \rightarrow N, x \mapsto x^p$. Since N is abelian, ϕ_p is in fact a homomorphism (why?). Observe that since $\forall g \in G, \phi_p(gxg^{-1}) = (gxg^{-1})^p = gx^pg^{-1} = g\phi_p(x)g^{-1}$, so in fact, $\ker \phi_p \trianglelefteq G$. Again, using the fact that N is a minimal (nontrivial) normal subgroup of G , and $\ker \phi_p \leq N$, we see that $\ker \phi_p = 1$ or N . Choose $p \mid |N|$, then by Cauchy's theorem, $\exists x \in N$ such that $\text{ord}(x) = p$. This forces x to be a non-trivial element of $\ker \phi_p$, so in fact, $\ker \phi_p = N$. Equivalently, $\forall x \in N, x^p = e$.

The last condition forces N to be a $\mathbb{Z}/p\mathbb{Z}$ -vector space with the scalar multiplication $k \cdot x = x^k$. Hence, we have $N \cong (\mathbb{Z}/p\mathbb{Z})^n$ as desired. $\square$

The following is a proposition given as qual problems in various years.

Proposition 9. Let G be a group generated by two elements a, b of order 2. Then, G is solvable.

Proof. Set $N = \langle ab \rangle$.

Claim. $N \trianglelefteq G$

Proof of the Claim. We have $N \trianglelefteq G \iff a, b \in N_G(N) \iff a(ab)a^{-1}, b(ab)b^{-1} \in N$. But the latter is true since $a(ab)a^{-1} = ba = (ab)^{-1} \in N$, and $b(ab)b^{-1} = ba = (ab)^{-1} \in N$. $\square$

Consider G/N . Since $abN = N$, G/N is in fact, a cyclic group generated by aN , so clearly, is abelian. Hence, $G^{(1)} = [G, G] \leq N$. But since N itself is abelian (as a cyclic group generated by ab), $G^{(2)} \leq [N, N] = \{e\}$. $\square$

Recall the following definitions and facts before moving onto one of the qual problems.

Definition 19 (Outer Automorphism Group). Recall that given a group H , $\text{Inn}(H) \trianglelefteq \text{Aut}(H)$. The *outer automorphism group* is defined as the quotient $\text{Out}(H) := \text{Aut}(H)/\text{Inn}(H)$.

Theorem 17 (Schreier Conjecture). If H is a finite simple group, then $\text{Out}(H)$ is solvable.

Fall 2021 Problem 9: Use Schreier conjecture to show that if G is a finite group such that both N and G/N are simple non-abelian, then $G \cong N \times G/N$.

Proof. We will construct a left splitting map $s : G \rightarrow N$ of the inclusion $N \hookrightarrow G$; the claim then, follows from theorem 13.

Consider $G \xrightarrow{\phi} \text{Aut}(N) \xrightarrow{\pi} \text{Out}(N) = \text{Aut}(N)/\text{Inn}(N)$ where $\phi : g \mapsto \sigma_g : n \mapsto gng^{-1}$, and π is the projection map.

By the first isomorphism theorem, we have the induced homomorphism $G/\ker(\pi \circ \phi) \hookrightarrow \text{Out}(N)$. Note by definition, $\forall n \in N, \phi(n) \in \text{Inn}(N)$, so $N \leq \phi^{-1}(\text{Inn}(N)) = \ker(\pi \circ \phi)$. So since G/N is simple, and by the correspondence between normal subgroups of G containing N and normal subgroups of G/N , $\ker(\pi \circ \phi)$ is either N or G . If $\ker(\pi \circ \phi) = N$, then we would have an embedding of a non-solvable group G/N into a solvable group $\text{Out}(N)$, a contradiction (this is where we use $\text{Out}(N)$ is solvable). Hence, $\ker(\pi \circ \phi) = G$, and so we have $\phi : G \twoheadrightarrow \text{Inn}(N)$.

Next, consider the restriction of ϕ to $N \trianglelefteq G$. Then, we have $\phi|_N : N \twoheadrightarrow \text{Inn}(N)$. As before, since N is simple, $\phi|_N$ is either N or $\{e\}$. We cannot have $\phi|_N = N$, because then, it'll force N to be abelian. Hence, $\phi|_N = \{e\}$ and we in fact, have an isomorphism $\phi|_N : N \xrightarrow{\cong} \text{Inn}(N)$.

From here, define $s : G \rightarrow N$ to be $\phi|_N^{-1} \circ \phi$, which is clearly an identity on N . $\square$

Extra Material: Category Theory Qual Problems

In this extra material section, we will go over the basics of Category theory and solve some qual problems involving in it. The topics to be covered are definitions of categories (objects, morphisms, as well as conditions on morphisms), functors and natural transformations, and (co)limits.

Definition 20 (Category). A category $\mathcal{C}$ consists of a following data:

- A class of objects $\text{Obj}(\mathcal{C})$.
- A class of morphisms $\text{Mor}(\mathcal{C})$, $f : A \rightarrow B$ between the objects $A, B \in \text{Obj}(\mathcal{C})$.
- A composition operation $A \xrightarrow{g} B \xrightarrow{f} C$, $f, g \mapsto f \circ g$.

Which satisfies the following conditions:

- (Identity) Every object A has a morphism $\text{id}_A : A \rightarrow A$, called an *identity*, which is the identity with respect to composition: $\forall f : A \rightarrow B$, $\text{id}_B \circ f = f \circ \text{id}_A = f$. It's an exercise that indeed, identity morphisms are unique.
- (Associativity) For all arrows of the form $A \xrightarrow{h} B \xrightarrow{g} C \xrightarrow{f} D$, $(f \circ g) \circ h = f \circ (g \circ h)$.

Example 9. Below are examples of categories.

- (I) **Set** is the category where objects are all sets and morphisms are functions between two sets. The composition is the usual composition.
- (II) **Top** is the category whose objects are topological spaces and morphisms are continuous functions $f : X \rightarrow Y$ between two topological spaces X and Y .
- (III) **Top_{*}** is the category of pointed topological spaces. The objects are (X, p) , a topological space X together with a chosen point $p \in X$ and morphisms between $(X, p), (Y, q)$ are continuous functions $f : X \rightarrow Y$ such that $f(p) = q$.
- (IV) **Grp** is the category of all groups and whose morphisms are homomorphisms between two groups.
- (V) **Ab** is the category of all abelian groups whose morphisms are homomorphisms between two groups.
- (VI) Let F be a field. Then $F - \mathbf{vekt}$ is the category of all F -vector spaces and morphisms are F -linear maps between two vector spaces.
- (VII) **Poset** is the space of all posets and morphisms are (non-strictly) monotone increasing functions. That is, a morphism between $(P, \leq)$ and $(Q, \leq)$ consists of functions $f : P \rightarrow Q$ where $\forall x, y \in P, x \leq y \implies f(x) \leq f(y)$.
- (VIII) Let P be a poset. Then, P itself is a category whose objects are elements $x \in P$ and for each $x \leq y$, we have a single morphism $x \rightarrow y$ if $x \leq y$, and no morphism $x \rightarrow y$ otherwise. For instance, for a topological space X , the set of all open sets $\text{Open}(X)$ (with $U \leq V \iff U \supseteq V$, note the reverse inclusion!) is very important in sheaf theory. Every poset P is a small category.
- (IX) **Ring** is the space of all rings with morphisms $A \xrightarrow{f} B$ between A and B to be ring homomorphisms (we will define more precisely what a ring homomorphism is).

Definitions and Facts 3 (Category Theory Basics). Let $\mathcal{C}$ be a category.

- (I) (Isomorphism) A morphism $f : A \rightarrow B$ is an *isomorphism* if there exists (necessarily unique) $g : B \rightarrow A$ such that $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$.
- (II) A category is called *locally small* if for each objects A, B $\text{Hom}_{\mathcal{C}}(A, B)$ is a set.
- (III) A category is called *small* if the class of objects and morphisms are small.
- (IV) Given a category $\mathcal{C}$, an opposite category $\mathcal{C}^{\text{op}}$ is the category with same objects, but with arrows (morphisms) and compositions reversed.
- (V) Let $\mathcal{C}$ be a category. We say a morphism $A \xrightarrow{f} B$ is a *monomorphism* if for any two morphisms $X \xrightarrow[a_2]{a_1} A$, if $f \circ a_1 = f \circ a_2$, then $a_1 = a_2$. Dually, we say $A \xrightarrow{f} B$ is an *epimorphism* if for any morphisms $B \xrightarrow[g_2]{g_1} Y$, $g_1 \circ f = g_2 \circ f \implies g_1 = g_2$.

Example 10. Here are some examples of monomorphisms and epimorphisms in various categories:

- (I) In **Set**, the monomorphisms are exactly injective functions (consider $\{*\} \rightarrow A$), and the epimorphisms are exactly the surjective functions $f : A \twoheadrightarrow B$ (consider the indicator function $\chi_{f(A)} : B \rightarrow \{0, 1\}$ and the constant function $B \rightarrow \{1\}$).
- (II) In **Top**, the monomorphisms are exactly the injective continuous functions (note every function $\{*\} \rightarrow A$ from a one-pointed space is continuous).
- (III) In **Ab**, the monomorphisms are exactly the injective homomorphisms (look at $\mathbb{Z} = F(a) \rightarrow A$); the epimorphisms are exactly the surjective homomorphisms (exercise!)
- (IV) The same holds true in **Grp**. The harder part, that epimorphisms in **Grp** are exactly the surjective homomorphisms, will be proven later.
- (V) In **Haus**, the space of all Hausdorff spaces (with morphisms continuous functions), the inclusion map $i : \mathbb{Q} \rightarrow \mathbb{R}$ is both a monomorphism and an epimorphism, even though, i is far from surjective! For the proof, use the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$ and that for any two $f, g : \mathbb{R} \rightarrow X$, the subspace $\{x \in \mathbb{R} \mid f(x) = g(x)\}$ is a closed set (this is where we use X is Hausdorff!).
- (VI) In **Poset**, take two partial orders $\leq_P, \leq'_P$ on a set P where $\leq'_P$ strictly extends $\leq_P$. Then, the identity map $\text{id} : P \rightarrow P$ (seen as a morphism from $(P, \leq_P)$ to $(P, \leq'_P)$) is a bijective map and hence, is both a monomorphism and an epimorphism, but the inverse (of the identity as a function) is not a morphism!
- (VII) In fact, in any given poset P , any two morphisms $x \leq y$ is both a monomorphism and an epimorphism. But unless $x = y$, x cannot be isomorphic to y .

Remark. As we saw in many examples, while it is true that an isomorphism $f : A \xrightarrow{\cong} B$ is both a mono and an epi, the converse need not hold true! But the converse does hold true if we strengthen our assumption.

Definition 21 (Split Epimorphism). We say a morphism $A \xrightarrow{r} B$ is a *split-epimorphism* if there exists a section (right inverse) $s : B \rightarrow A$ such that $r \circ s = \text{id}_B$. We also call r a *retract*.

Remark. As the name suggests, it's not hard to check r is an epimorphism (exercise!).

Proposition 10. Let $f : A \rightarrow B$ be both a split epimorphism and a monomorphism. Then, f is an isomorphism.

Proof. Let $s : B \rightarrow A$ be a right inverse, so that $f \circ s = \text{id}_B$. We also want to show $s \circ f = \text{id}_A$. By precomposing both sides of the first equation with f , we get $f \circ s \circ f = f = f \circ \text{id}_A$. But since f is a monomorphism, $s \circ f = \text{id}_A$. $\square$

We now prove that every epimorphisms in **Grp** are exactly the surjective homomorphisms.

Proposition 11. Let $f : G \rightarrow H$ be a homomorphism. If f is an epimorphism, then f is surjective.

Proof. Consider the left cosets $H/f(G)$. We have $H \curvearrowright H/f(G)$ by left-multiplication $h \cdot kf(G) = hkf(G)$, which as before, induces a homomorphism $\alpha : H \rightarrow S_{H/f(G)}$ given by $\alpha(h)(kf(G)) = hkf(G)$. Pick a point $p \notin H/f(G)$, and consider $X = H/f(G) \cup \{p\}$. Then, clearly, $S_{H/f(G)} \hookrightarrow S_X$, so we also have a homomorphism $\alpha : H \rightarrow S_X$ given by composing α with the embedding.

Consider the 2-cycle $\sigma = (p \ f(G)) \in S_X$, and the conjugation by σ , $\phi_\sigma : S_X \xrightarrow{\text{iso}} S_X, \tau \mapsto \sigma\tau\sigma^{-1}$. Composing ϕ_σ gives us another homomorphism $H \xrightarrow{\phi_\sigma \circ \alpha} S_X$ which corresponds to the action $H \curvearrowright X$ given by $h \cdot x = \sigma(h \cdot \sigma^{-1}(x))$ (why?).

Now, we have two maps $H \xrightarrow{f} G \xrightarrow[\phi_\sigma \circ \alpha]{\alpha} S_X$ α and $\phi_\sigma \circ \alpha$.

Claim. $\phi_\sigma \circ \alpha \circ f = \alpha \circ f$.

Proof of the Claim. For all $g \in G$, $\phi_\sigma \circ \alpha(f(g)) = \sigma \circ \alpha(f(g)) \circ \sigma^{-1}$. But note $\alpha(f(g))$ must fix both $f(G)$ and p , and hence, commutes with $\sigma = (f(G) \ p)$. Hence, this $\sigma \circ \alpha(f(g)) \circ \sigma^{-1} = \alpha(f(g))$, which we what we wanted to show. $\square$

Since f is an epimorphism, it follows $\phi_\sigma \circ \alpha = \alpha$. In particular, $\forall h \in H$, $\phi_\sigma \circ \alpha(h) = \sigma \circ \alpha(h) \circ \sigma^{-1} = \alpha(h)$. Hence, $\alpha(h)$ also commutes with $\sigma = (f(G) \ p)$, and since $\alpha(h)$ fixes p , it must also fix $f(G)$ as well. But that means, $\alpha(h)(f(G)) = hf(G) = f(G)$, or equivalently, $h \in f(G)$. Since $h \in H$ was arbitrary, it follows $H = f(G)$. $\square$

Next, we will move onto functors and natural transformations.

Definition 22. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ from the category $\mathcal{C}$ to $\mathcal{D}$ consists of the following data:

- A function class $F : \text{Obj}(\mathcal{C}) \rightarrow \text{Obj}(\mathcal{D})$ sending an object $A \in \text{Obj}(\mathcal{C})$ to $F(A) \in \text{Obj}(\mathcal{D})$
- A function class $F : \text{Mor}(\mathcal{C}) \rightarrow \text{Mor}(\mathcal{D})$ sending a morphism $A \xrightarrow{f} B$ to $FA \xrightarrow{F(f)} FB$

$$\begin{array}{ccc} A & \xrightarrow{F} & FA \\ f \downarrow & \xrightarrow{F} & \downarrow Ff \\ B & \xrightarrow{F} & FB \end{array}$$

Which satisfies the following:

- $F(\text{id}_A) = \text{id}_{FA}$
- $F(f \circ g) = F(f) \circ F(g)$

Definition 23. Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$ be two functors. A natural transformation $\alpha : F \Rightarrow G$ is a class of morphisms $\alpha_X : FX \rightarrow GX$ which is natural in X . That is, for any $X \xrightarrow{f} Y$, the following commutative diagram commutes:

$$\begin{array}{ccc} FX & \xrightarrow{\alpha_X} & GX \\ \downarrow Ff & & \downarrow Gf \\ FY & \xrightarrow{\alpha_Y} & GY \end{array}$$

Definition 24. Let I be a small category, and let $X : I \rightarrow \mathcal{C}$ be a functor (and we'll denote $X_i = X(i)$).

- (I) A *cone* over X is an object $C \in \mathcal{C}$ together with morphisms $c_i : C \rightarrow X_i$ such that $\forall i \xrightarrow{\alpha} j, c_j = X\alpha \circ c_i$. This corresponds to a natural transformation between the diagonal functor $\Delta C : I \rightarrow \mathcal{C}, i \mapsto C, (i \xrightarrow{\alpha} j) \mapsto (C \xrightarrow{\text{id}_C} C)$ and X .

$$\begin{array}{ccc} & & X_i \\ & \nearrow c_i & \downarrow X\alpha \\ C & & \\ & \searrow c_j & \downarrow \\ & & X_j \end{array}$$

- (II) A limit of X , denoted $\varprojlim_i X_i$ is a final object of category of all cones of over X . Unpacked, $\varprojlim_i X_i$ (together with the projection maps $p_i : \varprojlim_i X_i \rightarrow X_i$) satisfy the following universal property: $\forall C$ and a class of maps $d_i : C \rightarrow X_i$ such that $\forall \alpha : i \rightarrow j, c_j = X\alpha \circ c_i$, there exists unique $f : C \rightarrow \varprojlim_i X_i$ such that $c_i = p_i \circ f$.

$$\begin{array}{ccccc} & & c_i & \nearrow & X_i \\ & & & & \downarrow X\alpha \\ C & \xrightarrow{\exists! f} & \varprojlim_i X_i & \xrightarrow{p_i} & \\ & & & & \downarrow \\ & & c_j & \searrow & X_j \end{array}$$

Remark. Similarly, we can dualize the notion of cone to a cocone, and limit to a colimit. In fact, cocone of $X : I \rightarrow \mathcal{C}$ is a cone of $X^{\text{op}} : I^{\text{op}} \rightarrow \mathcal{C}^{\text{op}}$ and colimit of X is the limit of X^{op} .

Below are two prototypical examples of limits in a category.

Example 11 (Prototypical Examples of Limits). Let $\mathcal{C}$ be a (locally-small) category, I be a small category, and $X : I \rightarrow \mathcal{C}$ a functor.

- (I) If I is discrete, i.e. has no arrows other than identity, the limit is precisely the product $\prod_{i \in I} X_i$, together with the projection maps $\pi_i : \prod_{i \in I} X_i \rightarrow X_i$, if it exists. When I has two objects, the product is the usual product of two objects in a category.
- (II) If I is the diagram $\cdot \rightrightarrows \cdot$ (and X sends the objects and arrows to say,

$$A \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} B \text{), then the limit is precisely the equalizer } E \xrightarrow{j} A \text{ of } f \text{ and } g.$$

Remark. In **Set**, **Top**, **Grp**, **Ab**, **Poset**, etc., the product $\prod_{i \in I} X_i$ is a set of all $(x_i)_{i \in I}$ with $x_i \in X_i$, together with the appropriate structure. In **Top**, it is given the product topology; in **Grp**, **Ab**, the group operation is given coordinate-wise; and in **poset**, comparison is given coordinate-wise as well.

The equalizers are the subset $E = \{a \in A \mid f(a) = g(a)\}$, again, together with the appropriate structure. In **Top**, the space is given a subspace topology; in **Grp** and **Ab**, the group operation is induced from A ; and in **Poset**, the comparison relation is also induced from A as well.

In fact, as we will see, every limit is derived from these two types of limits.

Remark. Let I be a small category, and $X : I \rightarrow \mathcal{C}$ a functor, and assume $\mathcal{C}$ admits I -products. Let $C \in \mathcal{C}$. Then, an indexed collection of maps $c_i : C \rightarrow X_i$, $(c_i)_{i \in I}$ is in one-to-one correspondence with the map $c : C \rightarrow \prod_{i \in I} X_i$, given by $c \mapsto (\pi_i \circ c)_{i \in I}$.

Now, consider two maps $\prod_{i \in I} X_i \xrightarrow[\psi]{\phi} \prod_{\alpha: i \rightarrow j} X_j$ with ϕ given by $\pi_\alpha \circ \phi = \pi_j$ and $\pi_\alpha \circ \psi = X_\alpha \circ \pi_i$. Then, the condition that for all $\alpha : i \rightarrow j$, $c_j = X_\alpha \circ c_i$ is equivalent to the condition that $\phi \circ c = \psi \circ c$. This proves the following:

Proposition 12. Let I be a small category, let and $X : I \rightarrow \mathcal{C}$ be a functor. If $\mathcal{C}$ admits an I -product, then, $\varprojlim_i X_i$ (together with the projection maps $p_i : \varprojlim_i X_i \rightarrow X_i$) is given as the equalizer $p = (p_i)_{i \in I} : \varprojlim_i X_i \rightarrow \prod_i X_i$ of the two maps ϕ and ψ described in the remark.

$$\varprojlim_i X_i = \text{eq}(\phi, \psi) \xrightarrow{p=(p_i)_{i \in I}} \prod_{i \in I} X_i \xrightarrow[\psi=(X_\alpha \circ \pi_i)_\alpha]{\phi=(\pi_j)_\alpha} \prod_{\alpha: i \rightarrow j} X_j$$

Corollary 6. A category $\mathcal{C}$ is complete (admits all limits) if and only if it admits all products and equalizers. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ preserves limits if and only if it preserves products and equalizers.

Remark. Dually, we can construct every colimit as coequalizer of a product. We leave it up to you as an exercise to phrase the precise statement and to prove it.

Example 12. In **Set**, the limit of $X : I \rightarrow \mathbf{Set}$ is in fact given by

$$\varprojlim X_i = \{(x_i)_{i \in I} \mid x_i \in X_i, X_\alpha(x_i) = x_j \text{ for all } i \xrightarrow{\alpha} j\}.$$

We leave it up to you as an exercise to figure out and prove what limits are in various categories (given by sets with extra "structure").

Fall 2020 Problem 10: Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and suppose that every pair of morphisms in $\mathcal{C}$ admits a coequalizer. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor that preserves coequalizers: i.e., if $f, g : A \rightarrow B$ are morphisms in $\mathcal{C}$ and $\pi : B \rightarrow \text{coeq}(f, g)$ is the coequalizer morphism, then $F(\pi)$ is the coequalizer morphism for $F(f)$ and $F(g)$. Suppose also that if h is a morphism in $\mathcal{C}$ such that $F(h)$ is an isomorphism, then h is an isomorphism. Show that F is faithful.

Proof. Let $f, g : A \rightarrow B$ be two morphisms such that $F(f) = F(g)$. We want to show $f = g$. To that end, consider the coequalizer of f, g as shown below, and look at what F does to that diagram:

$$A \xrightarrow[f]{g} B \xrightarrow{\pi} \text{coeq}(f, g) \quad F A \xrightarrow[Ff=Fg]{Ff=Fg} F B \xrightarrow[F\pi]{F\pi} F \text{coeq}(f, g)$$

So we have $F B \xrightarrow{F\pi} F \text{coeq}(f, g)$ is a coequalizer of equal morphisms $Ff = Fg$. Since $\text{id}_{FB} : F B \rightarrow F B$ is such that $\text{id}_{FB} \circ Ff = \text{id}_{FB} \circ Fg$ (i.e. $Ff = Fg$), by the universal

property of coequalizers, there exists a unique arrow $h : F\text{coeq}(f, g) \rightarrow FB$ such that $h \circ F\pi = \text{id}_{FB}$.

$$\begin{array}{ccc} FA & \xrightarrow{Ff=Fg} & FB \\ & \searrow \text{id}_{FB} & \downarrow \exists! h \\ & & FB \end{array}$$

So $F\pi$ is a split monomorphism (dual to split epimorphism) which is also an epimorphism, and hence, $F\pi$ must in fact be an isomorphism. But then, by assumption, π is an isomorphism. Since by construction, $\pi \circ f = \pi \circ g$, composing both sides with π^{-1} , we get $f = g$. $\square$

Fall 2021 Problem 4: Consider a sequence of sets S_i for $i \geq 0$ and maps $\phi_i : S_i \rightarrow S_{i-1}$ for $i \geq 1$. Suppose that there exists a positive integer N such that the orders of the images of the maps ϕ_i are bounded above by N . Show that $\varprojlim_i S_i$ is finite.

Proof. The limit $\varprojlim_i S_i$ can be described as the set

$$S = \{(s_i)_{i \geq 0} \mid \phi(s_i) = s_{i-1} \text{ for all } i \geq 1\}.$$

We will show that $|S| \leq N$.

Claim. Suppose $s \neq t \in S$. Then, there exists i such that $\forall j \geq i, s_j \neq t_j$.

Proof of the Claim. First, choose i such that $s_i \neq t_i$ (such an i must exist since $s \neq t$). Then, note $s_i = \phi_{i+1}(s_{i+1})$ and $t_i = \phi_{i+1}(t_{i+1})$. So since $\phi_{i+1}(s_{i+1}) \neq \phi_{i+1}(t_{i+1})$, $s_{i+1} \neq t_{i+1}$. By induction on j , we can show that for all $j \geq i$, $s_j \neq t_j$. $\square$

Suppose now, we had $N + 1$ distinct elements $s^{(1)}, s^{(2)}, \dots, s^{(N)}, s^{(N+1)} \in S$. For each pair $k \neq m$, choose $i := i(k, m) = i(m, k)$ such that for all $j \geq i$, $s_j^{(k)} \neq s_j^{(m)}$. Set $i = \max(\{i(m, k) \mid m \neq k\})$. Then, $s_i^{(1)}, s_i^{(2)}, \dots, s_i^{(N)}, s_i^{(N+1)}$ are all distinct in S_i . But since $s_i^{(m)} = \phi_{i+1}(s_{i+1}^{(m)})$ for all $m = 1, 2, \dots, N, N + 1$, we have $N + 1$ distinct $s_i^{(m)}$'s in $\text{Im}(\phi_{i+1})$, contradicting that $|\text{Im}(\phi_{i+1})| \leq N$. $\square$

We'll give some examples of colimits in various categories.

Example 13. Let $X : I \rightarrow \mathcal{C}$ be a functor with $\mathcal{C}$ various possible categories.

- In **Set**, the coproduct of X_i 's is $\bigsqcup_{i \in I} X_i = \{(x, i) \mid x \in X_i\}$ the disjoint union of X_i 's. In **Top**, **Poset**, it is given the disjoint union topology (U is open iff $U \cap X_i$ is open in X_i) and partial order induced from each X_i (with no further relations).
- In **Grp**, coproduct of X_i 's is exactly the free product $\star_{i \in I} X_i$ (a set of words in X_i 's with appropriate relations, etc.). In **Ab**, $F\text{-vekt}$, the coproduct of X_i 's is exactly the direct sum $\oplus_{i \in I} X_i = \{(x_i) \mid x_i \in X_i, x_i \neq 0 \text{ for only finitely many } i \in I\}$ (with the pointwise operations).
- In **Ab**, coequalizer of $A \xrightleftharpoons[g]{f} B$ is exactly $B/\text{Im}(f - g)$ (together with the quotient map $\pi : B \twoheadrightarrow B/\text{Im}(f - g)$).

With this, we can tackle one of the qual problems on construction of colimits in **Ab**.

Spring 2024 Problem 10: Let F be a functor from a small category I to the category of abelian groups. Show that F admits a colimit by constructing it as a quotient of $\oplus_{i \in I} F(i)$, where i runs over the set of objects of I .

Proof. Let $f_i : F(i) \hookrightarrow \oplus_{i \in I} F(i)$ be the canonical injections ($F(i) \ni x \mapsto f_i(x) = (x_j)_{j \in I}$ where $x_j = 0$ if $j \neq i$, and $x_i = x$). As we have stated, the colimit of F is the coequalizer of $\oplus_{i \in I} F(i) \xrightarrow[(f_j \circ F\alpha)]{(f_i)} \oplus_{i \in I} F(i)$. Hence, by this is $\oplus_{i \in I} F(i) / \text{Im}((f_i) - (f_j \circ F\alpha))$. Note the image of $(f_i) - (f_j \circ F\alpha)$ is generated by $f_i(x) - f_j \circ F\alpha(x)$ as $x \in F(i)$, and $i \in I$. So the colimit is the quotient group of $\oplus_{i \in I} F(i)$ under the subgroup generated by $f_i(x) - f_j \circ F\alpha(x)$, $x \in F(i)$, $i \in I$. $\square$

Discussion 10 - Adjoint and Yoneda Lemma

Today, we will go over qual problems involving adjoints. We will recall the definitions.

Definition 25. Let $L : \mathcal{C} \rightarrow \mathcal{D}, R : \mathcal{D} \rightarrow \mathcal{C}$ be two functors. Consider two functors

$$\begin{aligned} \text{Hom}_{\mathcal{D}}(L(-), (-)) : \mathcal{C}^{\text{op}} \times \mathcal{D} &\rightarrow \mathbf{Set} \\ (A, B) &\mapsto \text{Hom}_{\mathcal{D}}(LA, B) \end{aligned}$$

and

$$\begin{aligned} \text{Hom}_{\mathcal{C}}((-), R(-)) : \mathcal{C}^{\text{op}} \times \mathcal{D} &\rightarrow \mathbf{Set} \\ (A, B) &\mapsto \text{Hom}_{\mathcal{C}}(A, RB) \end{aligned}$$

We say $L \dashv R$ (or R is a right adjoint to L , or L is a left adjoint to R) if the two functors are (naturally) isomorphic.

Remark. The condition that the isomorphisms are natural is equivalent to being natural in each variables A and B . Unpacked, it means the following diagram commutes for all $A' \xrightarrow{g} A, B \xrightarrow{f} B'$:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{D}}(LA, B) & \xrightarrow{\cong} & \text{Hom}_{\mathcal{C}}(A, RB) \\ \downarrow f_* = f \circ (-) & & \downarrow Rf_* = Rf \circ (-) \\ \text{Hom}_{\mathcal{D}}(LA, B') & \xrightarrow{\cong} & \text{Hom}_{\mathcal{C}}(A, RB') \\ \downarrow (Lg)_* = (-) \circ Lg & & \downarrow g_* = (-) \circ g \\ \text{Hom}_{\mathcal{D}}(LA', B') & \xrightarrow{\cong} & \text{Hom}_{\mathcal{C}}(A', RB') \end{array}$$

(Note the precomposition goes backwards!)

One of the most helpful tools for solving adjoint problems (and in general, in category theory) is what's called the Yoneda lemma. First, recall the definition of the Yoneda embedding:

Definition 26 (Yoneda Embedding). Let $\mathcal{C}$ be a category. For each $X \in \mathcal{C}$, we can assign a functor $R_X := \text{Hom}_{\mathcal{C}}((-), X) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$, $A \mapsto \text{Hom}_{\mathcal{C}}(A, X)$. We call the assignment $X \mapsto R_X = \text{Hom}_{\mathcal{C}}((-), X)$ is called *Yoneda embedding*, and is denoted $\mathbf{y}$. For $f : X \rightarrow Y$, if we set $\mathbf{y}(f) = f_* = f \circ (-) : \text{Hom}_{\mathcal{C}}((-), X) \rightarrow \text{Hom}_{\mathcal{C}}((-), Y)$, then $\mathbf{y} : \mathcal{C} \rightarrow \mathbf{Fun}(\mathcal{C}^{\text{op}}, \mathbf{Set}) = \mathbf{Set}^{\mathcal{C}^{\text{op}}}$ is a functor. See diagram below:

$$\begin{array}{ccc} & & \mathbf{y} : \mathcal{C} \rightarrow \mathbf{Set}^{\mathcal{C}^{\text{op}}} \\ & & \\ X & \longmapsto & \text{Hom}_{\mathcal{C}}((-), X) \\ f \downarrow & \longmapsto & \downarrow f_* = f \circ (-) \\ Y & \longmapsto & \text{Hom}_{\mathcal{C}}((-), Y) \end{array}$$

Remark. Sometimes, the Yoneda embedding $\mathbf{y}$ is instead, denoted $\mathfrak{y} : \mathcal{C} \rightarrow \mathbf{Set}^{\mathcal{C}^{\text{op}}}$. This is the Japanese character pronounced "yo". We will most likely stick to $\mathbf{y}$.

We have not proven that the suggestibly named Yoneda embedding is an embedding. That's what Yoneda lemma shows. In fact, it shows something more general.

Theorem 18 (Yoneda Lemma). Let $\mathcal{C}$ be a locally small category, and let $\mathbf{y} : \mathcal{C} \rightarrow \mathbf{Set}^{\mathcal{C}^{\text{op}}}$ be the Yoneda embedding given by $\mathbf{y}X = \text{Hom}_{\mathcal{C}}((-), X)$. Then, for any contravariant functor $F : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ (and hence, an object of the category $\mathbf{Set}^{\mathcal{C}^{\text{op}}}$), the function (class) given by

$$\begin{aligned} \mathbf{Nat}(\text{Hom}_{\mathcal{C}}((-), X), F) &\rightarrow F(X) \\ \alpha &\mapsto \alpha_X(\text{id}_X) \end{aligned}$$

is a bijection.

And as a corollary of the more general fact applied to $F = \text{Hom}_{\mathcal{C}}((-), Y)$, we get that the Yoneda embedding $\mathbf{y}$ is indeed fully-faithful:

Corollary 7. The Yoneda embedding $\mathbf{y}$ is fully-faithful.

Proof. Check that the bijection $\mathbf{Nat}(\text{Hom}_{\mathcal{C}}((-), X), \text{Hom}_{\mathcal{C}}((-), Y)) \rightarrow \text{Hom}_{\mathcal{C}}((-), Y)(X) = \text{Hom}_{\mathcal{C}}(X, Y)$ given in the Yoneda lemma is the inverse of $f \mapsto \mathbf{y}(f) = f_* = f \circ (-)$. $\square$

Lastly, we will reproduce our results on adjoints and (co)limits.

Proposition 13. Let $L : \mathcal{C} \rightarrow \mathcal{D}$ and $R : \mathcal{D} \rightarrow \mathcal{C}$ be two functors which are adjoints ($L \dashv R$). Then, for any small category I , and functors $X : I \rightarrow \mathcal{C}$ and $Y : I \rightarrow \mathcal{D}$, $R \lim X_i \xrightarrow{\cong} \lim R X_i$, and $\text{colim} L Y_i \xrightarrow{\cong} L \text{colim} Y_i$.

Fall 2023 Problem 6: For a positive integer, let C_n be the category with objects $[1, n] := \{1, 2, \dots, n\}$ and morphisms $\text{Mor}(i, j)$ an empty set if $i > j$ and a singleton otherwise. For positive integers m and n , a nonstrictly increasing function $f : [1, n] \rightarrow [1, m]$ can be viewed as a functor $C_n \rightarrow C_m$. Prove that this functor f has right adjoint if and only if $f(1) = 1$.

Proof. Suppose we have a functor $R : C_m \rightarrow C_n$. Then, R is a right adjoint of f if and only if for all i, j , $\text{Hom}_{C_m}(f(i), j) \cong \text{Hom}_{C_n}(i, R(j))$. This is equivalent to $\forall i, j$, $f(i) \leq j \iff i \leq R(j)$ (why?), which in turn, is equivalent to $\forall i, j$, $i \in f^{-1}([1, j]) \iff i \leq R(j)$. Fix a $j \in C_m$. Then, $i \in f^{-1}([1, j])$ if and only if $i \leq R(j)$, and hence, $f^{-1}([1, j]) = [1, R(j)]$, and thus, $R(j) = \max f^{-1}([1, j])$.

So, we showed that R is a right adjoint if and only if $R(j) = \max f^{-1}([1, j])$ for all $j \in C_m$. Therefore, the right adjoint R is well-defined (exists) if and only if each $f^{-1}([1, j])$ is non-empty, but this is equivalent to $f^{-1}(\{1\}) \neq \emptyset$. But since f is monotone, the latter is equivalent to $f(1) = 1$, and we're done. $\square$

Spring 2023 Problem 1: Let $F, F' : \mathcal{C} \rightarrow \mathcal{D}$ and $G, G' : \mathcal{D} \rightarrow \mathcal{C}$ be four functors such that F is left adjoint to G and F' is left adjoint to G' . Establish a bijection between natural transformations $\alpha : F \Rightarrow F'$ and the natural transformations $\beta : G' \Rightarrow G$. [Hint: Use $G\alpha G' : GFG' \Rightarrow GF'G'$.]

Proof. We're defining a bijection

$$\begin{aligned} \mathbf{Nat}(F, F') &\rightarrow \mathbf{Nat}(G'G) \\ (\alpha : F \Rightarrow F') &\mapsto (\beta : G' \Rightarrow G) \end{aligned}$$

Consider any objects $X \in \mathcal{C}$ and $Y \in \mathcal{D}$, and $FX \xrightarrow{\alpha_X} F'X$. Then, we have the following diagram below:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{D}}(FX, Y) & \xrightarrow{\cong} & \text{Hom}_{\mathcal{C}}(X, GY) \\ \uparrow \alpha_X^* = (-) \circ \alpha_X & & \uparrow \\ \text{Hom}_{\mathcal{D}}(F'X, Y) & \xrightarrow{\cong} & \text{Hom}_{\mathcal{C}}(X, G'Y) \end{array}$$

The map on the right column is induced from the isomorphisms and the map on the left, $(-) \circ \alpha_X$.

Now, take any $X \xrightarrow{f} X'$. Then, we have the following two diagrams:

$$\begin{array}{ccc} \text{Hom}_{\mathcal{D}}(FX', Y) & \xrightarrow{(-) \circ Ff} & \text{Hom}_{\mathcal{D}}(FX, Y) & \quad & \text{Hom}_{\mathcal{C}}(X', GY) & \xrightarrow{(-) \circ f} & \text{Hom}_{\mathcal{C}}(X, GY) \\ (-) \circ \alpha_{X'} \uparrow & & (-) \circ \alpha_X \uparrow & & \uparrow & & \uparrow \\ \text{Hom}_{\mathcal{D}}(F'X', Y) & \xrightarrow{(-) \circ F'f} & \text{Hom}_{\mathcal{D}}(F'X, Y) & \quad & \text{Hom}_{\mathcal{C}}(X', G'Y) & \xrightarrow{(-) \circ f} & \text{Hom}_{\mathcal{C}}(X, G'Y) \end{array}$$

Since the left square commutes, and the dashed arrows are induced as to mirror the left square, the right square commutes as well. So, for any fixed $Y \in \mathcal{D}$, the dashed arrows form a natural transformation $\text{Hom}_{\mathcal{C}}((-), G'Y) \Rightarrow \text{Hom}_{\mathcal{C}}((-), GY)$. Hence, by Yoneda lemma, $\exists! \beta_Y \in \text{Hom}_{\mathcal{C}}(G'Y, GY)$ such that $\text{Hom}_{\mathcal{C}}((-), G'Y) \Rightarrow \text{Hom}_{\mathcal{C}}((-), GY)$ is given by $\beta_Y \circ (-)$.

Exercise 8. Check that $\beta_Y : G'Y \rightarrow GY$ form a natural transformation $\beta : G' \Rightarrow G$. For this, use the fact that $\text{Hom}_{\mathcal{D}}(FX, Y)$ and $\text{Hom}_{\mathcal{D}}(F'X, Y)$ are natural in the variable Y .

So we have an assignment $(\alpha : F \Rightarrow F') \mapsto (\beta : G' \Rightarrow G)$. We can use similar reasoning to construct an assignment $(\beta : G' \Rightarrow G) \mapsto (\alpha : F \Rightarrow F')$. Moreover, by definition, and

the uniqueness statement of Yoneda lemma, they are inverses of one another. Unpacked, the argument is as follows:

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{D}}(FX, Y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, GY) \\
(-) \circ \alpha_X \uparrow \uparrow (-) \circ \widetilde{\alpha}_X & & \uparrow \beta_Y \circ (-) \\
\mathrm{Hom}_{\mathcal{D}}(F'X, Y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, G'Y)
\end{array}
\qquad
\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{D}}(FX, Y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, GY) \\
(-) \circ \alpha_X \uparrow & & \beta_Y \circ (-) \uparrow \uparrow \widetilde{\beta}_Y \circ (-) \\
\mathrm{Hom}_{\mathcal{D}}(F'X, Y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, G'Y)
\end{array}$$

We have $(\alpha : F \Rightarrow F') \mapsto (\beta : G' \Rightarrow G) \mapsto (\tilde{\alpha} : F \Rightarrow F')$ (see the diagram on the left). Then, since both α and $\tilde{\alpha}$ make the diagram commute, $(-) \circ \alpha_X = (-) \circ \tilde{\alpha}_X$. By the uniqueness statement in the Yoneda lemma, $\alpha = \tilde{\alpha}$. Similarly, applying uniqueness statement of Yoneda lemma to on the right, $\beta = \tilde{\beta}$. $\square$

Remark. We haven't used the hint, and I don't know of a way to do this problem using the hint. If anyone knows of a way to do this using the hint, feel free to let me know.

In preparation for the next qual problem, we need a notion of unit and counit of the adjunction.

Theorem 19. Let $L : \mathcal{C} \rightarrow \mathcal{D}$ and $R : \mathcal{D} \rightarrow \mathcal{C}$ be two functors, with $L \dashv R$ (L is a left adjoint to R). Then, we have a way to induce natural transformations $\eta : \mathrm{Id} \Rightarrow RL$ and $\epsilon : LR \Rightarrow \mathrm{Id}$ called unit and counit of the adjunction. The canonical bijections are then, described as below:

$$\begin{array}{ccc}
& f \mapsto R(f) \circ \eta_X & \\
& \curvearrowright & \\
\mathrm{Hom}_{\mathcal{D}}(LX, Y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, RY) \\
& \curvearrowleft & \\
& g \mapsto \epsilon_Y \circ L(g) &
\end{array}$$

In particular, the unit and counit satisfy the *unit-counit identities* $\epsilon_{LX} \circ L\eta_X = \mathrm{Id}_{LX}$ and $R\epsilon_Y \circ \eta_{RY} = \mathrm{Id}_{RY}$:

$$\begin{array}{ccc}
LX & \xrightarrow{L\eta_X} & LRLX \xrightarrow{\epsilon_{LX}} LX \\
& \searrow \mathrm{Id}_{LX} & \nearrow \\
& &
\end{array}
\qquad
\begin{array}{ccc}
RY & \xrightarrow{\eta_{RY}} & RLR Y \xrightarrow{R\epsilon_Y} RY \\
& \searrow \mathrm{Id}_{RY} & \nearrow \\
& &
\end{array}$$

Proof. We define $\eta_X : X \rightarrow RLX$ and $\epsilon_Y : LRY \rightarrow Y$ as the maps corresponding to the identity maps $\mathrm{Id}_{LX} : LX \rightarrow LX$ and $\mathrm{Id}_{RY} : RY \rightarrow RY$ (see diagram!)

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{D}}(LX, LX) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, RLX) \\
\mathrm{Id}_{LX} \longmapsto & & \eta_X
\end{array}
\qquad
\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{D}}(LRY, Y) & \xleftarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(RY, RY) \\
\epsilon_Y \longleftarrow & & \mathrm{Id}_{RY}
\end{array}$$

Lemma 10. The morphisms η and ϵ are natural. That is, $RL(f) \circ \eta_X = \eta_Y \circ f$ and $\epsilon_Y \circ LR(g) = g \circ \epsilon_X$ (note the X, Y in the two different squares lie in different categories!)

$$\begin{array}{ccc}
X & \xrightarrow{\eta_X} & RLX \\
\downarrow f & & \downarrow RL(f) \\
Y & \xrightarrow{\eta_Y} & RLY
\end{array}
\qquad
\begin{array}{ccc}
X & \xleftarrow{\epsilon_X} & LRX \\
\downarrow g & & \downarrow LR(g) \\
Y & \xleftarrow{\epsilon_Y} & LRY
\end{array}$$

Proof of the Lemma. Consider the following diagrams:

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{D}}(LX, LX) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, RLX) & \quad & \mathrm{Hom}_{\mathcal{D}}(LRX, X) & \xleftarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(RX, RX) \\
\downarrow Lf \circ (-) & & \downarrow RLf \circ (-) & & \downarrow g \circ (-) & & \downarrow Rg \circ (-) \\
\mathrm{Hom}_{\mathcal{D}}(LX, LY) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, RLY) & \quad & \mathrm{Hom}_{\mathcal{D}}(LRX, Y) & \xleftarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(RX, RY) \\
\uparrow (-) \circ Lf & & \uparrow (-) \circ f & & \uparrow (-) \circ LRg & & \uparrow (-) \circ Rg \\
\mathrm{Hom}_{\mathcal{D}}(LY, LY) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(Y, RLY) & \quad & \mathrm{Hom}_{\mathcal{D}}(LRY, Y) & \xleftarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(RY, RY)
\end{array}$$

Follow Id_{LX} from the top left corner and Id_{LY} from the bottom left corner. They will both equal $L(f) \in \mathrm{Hom}_{\mathcal{D}}(LX, LY)$ in the center left, and end up being morphisms corresponding to $L(f)$ in $\mathrm{Hom}_{\mathcal{C}}(X, RLY)$. But following them via different arrows, we see that they must be sent to $RL(f) \circ \eta_X$ and $\eta_Y \circ f$. Hence, $RL(f) \circ \eta_X = \eta_Y \circ f$. Similarly, one can show that $g \circ \epsilon_X = \epsilon_Y \circ LR(g)$ $\square$

Lemma 11. The natural bijections are given as above. That is, $f \mapsto R(f) \circ \eta_X$ and $g \mapsto \epsilon_Y \circ L(g)$.

Proof of the Lemma. We chase diagrams again.

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathcal{D}}(LX, LX) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, RLX) \\
\downarrow f \circ (-) & & \downarrow Rf \circ (-) \\
\mathrm{Hom}_{\mathcal{D}}(LX, Y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(X, RY) \\
\uparrow (-) \circ Lg & & \uparrow (-) \circ g \\
\mathrm{Hom}_{\mathcal{D}}(LRX, Y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathcal{C}}(RY, RY)
\end{array}$$

Following where Id_{LX} from the top left corner lands in center right corner. This is where f (from center left) ends up in. But it must equal $R(f) \circ \eta_X$. Similarly, follow Id_{RY} from bottom right to the center left corner. This is the map corresponding to g and must equal $\epsilon_Y \circ L(g)$. $\square$

Lemma 12. The unit-counit relation holds: $\epsilon_{LX} \circ L\eta_X = \mathrm{Id}_{LX}$ and $R\epsilon_Y \circ \eta_{RY} = \mathrm{Id}_{RY}$:

$$\begin{array}{ccc}
LX & \xrightarrow{L\eta_X} & LRLX & \xrightarrow{\epsilon_{LX}} & LX \\
& \searrow & \text{Id}_{LX} & \nearrow & \\
RY & \xrightarrow{\eta_{RY}} & RLRY & \xrightarrow{R\epsilon_Y} & RY \\
& \searrow & \text{Id}_{RY} & \nearrow &
\end{array}$$

Proof of the Lemma. Consider the two diagrams showing natural bijections:

$$\mathrm{Hom}_{\mathcal{D}}(LX, LX) \xrightarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(X, RLX) \quad \mathrm{Hom}_{\mathcal{D}}(LRY, Y) \xleftarrow{\cong} \mathrm{Hom}_{\mathcal{C}}(RY, RY)$$

On the diagram to the left, we have $\mathrm{Id}_{LX} \mapsto \eta_X$ from left to right, and then, $\eta_X \mapsto \epsilon_{LX} \circ L\eta_X$ from right to left. But η_X must be sent back to Id_{LX} , so $\epsilon_{LX} \circ L\eta_X = \mathrm{Id}_{LX}$. Dually, we can show that $R\epsilon_Y \circ \eta_{RY} = \mathrm{Id}_{RY}$. $\square$

$\square$

Exercise 9. Conversely, suppose $L : \mathcal{C} \rightarrow \mathcal{D}$ and $R : \mathcal{D} \rightarrow \mathcal{C}$ are two functors. Show that if we're given natural transformations $\eta : \text{Id} \Rightarrow RL$ and $\epsilon : LR \Rightarrow \text{Id}$ satisfying the unit-counit identities, then $L \dashv R$. Moreover, the natural bijections are induced by η and ϵ and are described the same as before:

$$\begin{array}{ccc} & f \mapsto R(f) \circ \eta_X & \\ & \curvearrowright & \\ \text{Hom}_{\mathcal{D}}(LX, Y) & \xrightarrow{\cong} & \text{Hom}_{\mathcal{C}}(X, RY) \\ & \curvearrowleft & \\ & g \mapsto \epsilon_Y \circ L(g) & \end{array}$$

Fall 2018 Problem 7: Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor with a right adjoint G . Show that F is fully faithful if and only if the unit of the adjunction $\eta : \text{Id} \rightarrow GF$ is an isomorphism.

Proof. We'll do both parts simultaneously (up until near the end of the proof). Consider the diagram below

$$\begin{array}{ccc} \text{Hom}_{\mathcal{C}}(X, Y) & \xrightarrow{F} & \text{Hom}_{\mathcal{D}}(FX, FY) \\ \downarrow \eta_Y \circ (-) & \swarrow \cong & \\ \text{Hom}_{\mathcal{C}}(X, GFY) & & \end{array}$$

We will show that the diagram is commutative: that is, we show the map from $\text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{C}}(X, GFY)$ given by applying F and then, the natural bijection, is given by $f \mapsto \eta_Y \circ f$. Let $f : X \rightarrow Y$, then we get $f \mapsto Ff \mapsto GFf \circ \eta_X$. But since η is a natural transformation, $GFf \circ \eta_X = \eta_Y \circ f$.

Now, suppose $\eta : \text{Id} \Rightarrow GF$ is an isomorphism. Then, the left column is bijective, and hence, the map $\text{Hom}_{\mathcal{C}}(X, Y) \xrightarrow{F} \text{Hom}_{\mathcal{D}}(FX, FY)$ is bijective as well. This is exactly the condition that F is fully-faithful.

Conversely, suppose F is fully-faithful. Fix $Y \in \mathcal{C}$. Then, for all $X \in \mathcal{C}$, the map $\text{Hom}_{\mathcal{C}}(X, Y) \xrightarrow{\eta_Y \circ (-)} \text{Hom}_{\mathcal{C}}(X, GFY)$ is a natural isomorphism. Set β to be the inverse of the natural transformation $\eta_Y \circ (-) : \text{Hom}_{\mathcal{C}}((-), Y) \rightarrow \text{Hom}_{\mathcal{C}}((-), GFY)$. Then, by Yoneda lemma, there exists unique $\alpha_Y : GFY \rightarrow Y$ such that $\beta = \alpha_Y \circ (-)$. But then, α_Y and η_Y are inverses of one another (why?), so η itself is an isomorphism. $\square$

Fall 2022 Problem 9: Let $\phi : \mathcal{A}b \rightarrow \mathcal{G}p$ be the functor that takes an abelian group A to A in the category of groups. Show that ϕ has a left adjoint α . Does ϕ have a right adjoint? Does α have a left adjoint? Justify your answers.

Proof. The left adjoint of ϕ is given by the abelianization functor $\alpha : \mathcal{G}p \rightarrow \mathcal{A}b$

$$\begin{array}{ccc} G & \xrightarrow{\alpha} & G_{\text{ab}} \\ f \downarrow & \xrightarrow{\alpha} & \downarrow f_{\text{ab}} \\ H & \xrightarrow{\alpha} & H_{\text{ab}} \end{array}$$

We leave it up to you as an exercise to show this is a left adjoint of ϕ .

The functor ϕ does not have a right adjoint, since otherwise, ϕ would be a left adjoint, and ϕ must therefore, preserve colimits. The colimit of $\mathbb{Z}$ and $\mathbb{Z}$ in $\mathbf{Ab}$ is the direct sum

$\mathbb{Z} \oplus \mathbb{Z}$, but in **Grp**, is $\mathbb{Z} \star \mathbb{Z} \cong F(a, b)$. Hence, the canonical morphism $\mathbb{Z} \star \mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z}, a \mapsto (1, 0), b \mapsto (0, 1)$ cannot be an isomorphism.

Similarly, α does not have a left adjoint since α would have to preserve limits. Here, α *does* preserve products, but not equalizers. To see this, take any group G with $[G, G]$ nontrivial abelian (e.g. $G = \langle a, b \mid a^2 = b^2 = e \rangle$). Then, given the equalizer diagram below,

$$[G, G] \rightharpoonrightarrow G \begin{array}{c} \xrightarrow{\pi} \\ \xrightarrow[e=(g \mapsto e)]{} \end{array} G_{\text{ab}} = G/[G, G]$$

the functor α sends it to

$$[G, G] \rightharpoonrightarrow G_{\text{ab}} \begin{array}{c} \xrightarrow{\text{id}} \\ \xrightarrow[e]{} \end{array} G_{\text{ab}}$$

which is not the equalizer of the two arrows. In fact, the equalizer is $\{e\} \rightarrow G_{\text{ab}}$, and the induced map $[G, G]_{\text{ab}} = [G, G] \rightarrow \{e\}$ cannot be an isomorphism. $\square$

Remark. Indeed, the abelianization functor **Ab** $\rightarrow$ **Grp**, $G \mapsto G_{\text{ab}}$ does preserve co-products! That is, the map $(G \star H)_{\text{ab}} \cong G_{\text{ab}} \oplus H_{\text{ab}}$. We leave it up to you as an exercise to verify this manually.